

StataNow 19.5  
MP-Parallel Edition

## Statistics and Data Science

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Notes:

1. Unicode is supported; see [help unicode advice](#).
2. More than 2 billion observations are allowed; see [help obs advice](#).
3. Maximum number of variables is set to 5,000 but can be increased; see [help set maxvar](#).

```
1. use "/Users/michellekadiman/Library/CloudStorage/GoogleDrive-michellekadiman12@gmail.com/My Drive/00. My Pa
> ssport/12. NC State FM/06. 2026 Spring/FIM 601 Seminar in Financial Mathematics/10. Group Project/FIM601_CC
> AR/Macro_Panel_Data.dta"
```

2 . browse

```
3 . encode Entity, gen(entity_id)
```

```
4 . gen qdate = quarterly(Quarter, "YQ")
```

```
5 . format qdate %tq
```

```
6 . xtset entity_id qdate
```

Panel variable: **entity\_id** (strongly balanced)  
Time variable: **qdate**, 2000q1 to 2025q4  
Delta: **1 quarter**

7 . \* test dependency of delinquency rate to real GDP

```
8 . xtreg Delinquency_rate Real_GDP_growth, fe vce(cluster entity_id)
```

```
Fixed-effects (within) regression      Number of obs   =    206
Group variable: entity_id             Number of groups =     2
```

|                  |                |
|------------------|----------------|
| R-squared:       | Obs per group: |
| Within = 0.0133  | min = 103      |
|                  | avg = 103.0    |
| Overall = 0.0086 | max = 103      |

|                        |          |   |        |
|------------------------|----------|---|--------|
|                        | F(1, 1)  | = | 3.03   |
| corr(u_i, Xb) = 0.0000 | Prob > F | = | 0.3318 |

(Std. err. adjusted for 2 clusters in **entity\_id**)

|                  | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Delinquency_rate |             |                                   |       |       |                      |          |
| Real_GDP_growth  | -.0002647   | .000152                           | -1.74 | 0.332 | -.0021955            | .0016662 |
| _cons            | .0259826    | .0003479                          | 74.69 | 0.009 | .0215623             | .0304033 |
| sigma_u          | .01188763   |                                   |       |       |                      |          |
| sigma_e          | .01147948   |                                   |       |       |                      |          |
| rho              | .51746181   | (fraction of variance due to u_i) |       |       |                      |          |

```
9 . xtreg Delinquency_rate L(1/4).Real_GDP_growth, fe vce(cluster entity_id)
```

|                                   |                  |   |     |
|-----------------------------------|------------------|---|-----|
| Fixed-effects (within) regression | Number of obs    | = | 198 |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2   |

|                  |                |
|------------------|----------------|
| R-squared:       | Obs per group: |
| Within = 0.1936  | min = 99       |
|                  | avg = 99.0     |
| Overall = 0.1281 | max = 99       |

|                        |          |   |   |
|------------------------|----------|---|---|
| corr(u_i, Xb) = 0.0000 | F(1, 1)  | = | . |
|                        | Prob > F | = | . |

(Std. err. adjusted for 2 clusters in `entity_id`)

|                  | Robust      |           |   |      |                      |
|------------------|-------------|-----------|---|------|----------------------|
| Delinquency_rate | Coefficient | std. err. | t | P> t | [95% conf. interval] |

|                 |           |                                   |        |       |           |           |
|-----------------|-----------|-----------------------------------|--------|-------|-----------|-----------|
| Real_GDP_growth |           |                                   |        |       |           |           |
| L1.             | -.0005095 | .0001126                          | -4.52  | 0.139 | -.0019407 | .0009218  |
| L2.             | -.000691  | .000112                           | -6.17  | 0.102 | -.0021141 | .0007321  |
| L3.             | -.0007213 | .0000671                          | -10.75 | 0.059 | -.0015741 | .0001316  |
| L4.             | -.0005897 | .0000335                          | -17.63 | 0.036 | -.0010148 | -.0001647 |
| _cons           | .0307406  | .0007304                          | 42.09  | 0.015 | .0214602  | .0400211  |
| sigma_u         | .0117044  |                                   |        |       |           |           |
| sigma_e         | .01055697 |                                   |        |       |           |           |
| rho             | .55140711 | (fraction of variance due to u_i) |        |       |           |           |

```
10 . xtreg Delinquency_rate Real_GDP_growth L(1/4).Real_GDP_growth, fe vce(cluster entity_id)
```

|                                           |                  |   |             |
|-------------------------------------------|------------------|---|-------------|
| Fixed-effects (within) regression         | Number of obs    | = | 198         |
| Group variable: <b>entity_id</b>          | Number of groups | = | 2           |
| R-squared:                                | Obs per group:   |   |             |
| Within = <b>0.2314</b>                    | min =            |   | <b>99</b>   |
|                                           | avg =            |   | <b>99.0</b> |
| Overall = <b>0.1531</b>                   | max =            |   | <b>99</b>   |
|                                           | $F(1, 1)$        | = | .           |
| $\text{corr}(u_i, X_b) = \mathbf{0.0000}$ | Prob > F         | = |             |

(Std. err. adjusted for 2 clusters in **entity\_id**)

|                  | Robust      |                                   |        |       |                      |           |
|------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Delinquency_rate | Coefficient | std. err.                         | t      | P> t  | [95% conf. interval] |           |
| Real_GDP_growth  |             |                                   |        |       |                      |           |
| ---              | -.000457    | .0001997                          | -2.29  | 0.262 | -.0029947            | .0020807  |
| L1.              | -.0006128   | .0001581                          | -3.88  | 0.161 | -.0026212            | .0013957  |
| L2.              | -.0007404   | .0001339                          | -5.53  | 0.114 | -.0024416            | .0009608  |
| L3.              | -.0007405   | .0000757                          | -9.78  | 0.065 | -.0017021            | .0002212  |
| L4.              | -.0006105   | .0000426                          | -14.33 | 0.044 | -.0011518            | -.0000692 |
| _cons            | .0322069    | .0013731                          | 23.46  | 0.027 | .0147601             | .0496538  |
| sigma_u          | .0117044    |                                   |        |       |                      |           |
| sigma_e          | .01033311   |                                   |        |       |                      |           |
| rho              | .5619852    | (fraction of variance due to u_i) |        |       |                      |           |

```
11 . * test dependency of delinquency rate to nominal GDP
```

```
12 . xtreg Delinquency_rate Nominal_GDP_growth, fe vce(cluster entity_id)
```

|                                   |                  |   |               |
|-----------------------------------|------------------|---|---------------|
| Fixed-effects (within) regression | Number of obs    | = | 206           |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2             |
| R-squared:                        | Obs per group:   |   |               |
| Within = <b>0.0356</b>            | min =            |   | <b>103</b>    |
|                                   | avg =            |   | <b>103.0</b>  |
| Overall = <b>0.0232</b>           | max =            |   | <b>103</b>    |
|                                   | F(1, 1)          | = | <b>5.47</b>   |
| corr(u_i, Xb) = <b>0.0000</b>     | Prob > F         | = | <b>0.2573</b> |

(Std. err. adjusted for 2 clusters in **entity\_id**)

|                    |             | Robust                            |       |       |                      |          |
|--------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Delinquency_rate   | Coefficient | std. err.                         | t     | P> t  | [95% conf. interval] |          |
| Nominal_GDP_growth | -.0003768   | .0001612                          | -2.34 | 0.257 | -.002425             | .0016713 |
| _cons              | .0271489    | .0007581                          | 35.81 | 0.018 | .0175167             | .0367812 |
| sigma_u            | .01188763   |                                   |       |       |                      |          |
| sigma_e            | .01134871   |                                   |       |       |                      |          |
| rho                | .52318045   | (fraction of variance due to u_i) |       |       |                      |          |

```
13 . xtreg Delinquency_rate L(1/4).Nominal_GDP_growth, fe vce(cluster entity_id)
```

|                                   |                  |   |             |
|-----------------------------------|------------------|---|-------------|
| Fixed-effects (within) regression | Number of obs    | = | 198         |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2           |
| R-squared:                        | Obs per group:   |   |             |
| Within = <b>0.2099</b>            | min =            |   | <b>99</b>   |
|                                   | avg =            |   | <b>99.0</b> |
| Overall = <b>0.1389</b>           | max =            |   | <b>99</b>   |
|                                   | F(1, 1)          | = | .           |

corr(u\_i, Xb) = **0.0000** Prob > F = .  
(Std. err. adjusted for 2 clusters in **entity\_id**)

|                    | Robust      |                                   |        |       |                      |           |
|--------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Delinquency_rate   | Coefficient | std. err.                         | t      | P> t  | [95% conf. interval] |           |
| Nominal_GDP_growth |             |                                   |        |       |                      |           |
| L1.                | -.0003876   | .0000996                          | -3.89  | 0.160 | -.0016527            | .0008775  |
| L2.                | -.0005223   | .0000754                          | -6.92  | 0.091 | -.0014808            | .0004363  |
| L3.                | -.0005172   | .0000107                          | -48.41 | 0.013 | -.0006529            | -.0003814 |
| L4.                | -.0003842   | .0000297                          | -12.92 | 0.049 | -.0007622            | -6.26e-06 |
| _cons              | .0334881    | .0007198                          | 46.52  | 0.014 | .0243422             | .042634   |
| sigma_u            | .0117044    |                                   |        |       |                      |           |
| sigma_e            | .01044969   |                                   |        |       |                      |           |
| rho                | .55645442   | (fraction of variance due to u_i) |        |       |                      |           |

```
14 . xtreg Delinquency_rate Nominal_GDP_growth L(1/4).Nominal_GDP_growth, fe vce(cluster entity_id)
```

```
Fixed-effects (within) regression      Number of obs   =    198
Group variable: entity_id             Number of groups =     2
```

|                  |                |
|------------------|----------------|
| R-squared:       | Obs per group: |
| Within = 0.2345  | min = 99       |
|                  | avg = 99.0     |
| Overall = 0.1552 | max = 99       |

|                                     |                                   |   |   |
|-------------------------------------|-----------------------------------|---|---|
| <code>corr(u_i, Xb) = 0.0000</code> | $\frac{F(1, 1)}{\text{Prob} > F}$ | = | . |
|                                     |                                   | = | . |

(Std. err. adjusted for 2 clusters in **entity\_id**)

|                    |             | Robust                            |        |       |                      |           |
|--------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Delinquency_rate   | Coefficient | std. err.                         | t      | P> t  | [95% conf. interval] |           |
| Nominal_GDP_growth |             |                                   |        |       |                      |           |
| --.                | -.0003164   | .0001748                          | -1.81  | 0.321 | -.002538             | .0019052  |
| L1.                | -.0004174   | .0001163                          | -3.59  | 0.173 | -.001895             | .0010602  |
| L2.                | -.0004928   | .0000593                          | -8.31  | 0.076 | -.0012467            | .0002611  |
| L3.                | -.0004783   | .0000108                          | -44.42 | 0.014 | -.0006151            | -.0003415 |
| L4.                | -.0003708   | .0000373                          | -9.95  | 0.064 | -.0008444            | .0001029  |
| _cons              | .034724     | .0014046                          | 24.72  | 0.026 | .0168766             | .0525714  |
| sigma_u            | .0117044    |                                   |        |       |                      |           |
| sigma_e            | .01031245   |                                   |        |       |                      |           |
| rho                | .56297062   | (fraction of variance due to u_i) |        |       |                      |           |

15 . \* test dependency of delinquency rate to unemployment GDP

```
16 . xtreg Delinquency_rate Unemployment_rate, fe vce(cluster entity_id)
```

```
Fixed-effects (within) regression      Number of obs   =      206
Group variable: entity_id             Number of groups =       2
```

|                         |                    |
|-------------------------|--------------------|
| R-squared:              | Obs per group:     |
| Within = <b>0.1302</b>  | min = <b>103</b>   |
|                         | avg = <b>103.0</b> |
| Overall = <b>0.0847</b> | max = <b>103</b>   |

|                        |          |   |         |
|------------------------|----------|---|---------|
|                        | F(1, 1)  | = | 2436.01 |
| corr(u_i, Xb) = 0.0000 | Prob > F | = | 0.0129  |

(Std. err. adjusted for 2 clusters in **entity\_id**)

|                   |             | Robust                            |       |       |                      |          |
|-------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Delinquency_rate  | Coefficient | std. err.                         | t     | P> t  | [95% conf. interval] |          |
| Unemployment_rate | .002141     | .0000434                          | 49.36 | 0.013 | .0015898             | .0026922 |
| _cons             | .0132913    | .0002449                          | 54.28 | 0.012 | .01018               | .0164026 |
| sigma_u           | .01188763   |                                   |       |       |                      |          |
| sigma_e           | .01077789   |                                   |       |       |                      |          |
| rho               | .54884447   | (fraction of variance due to u_i) |       |       |                      |          |

```
17 . xtreg Delinquency_rate L(1/4).Unemployment_rate, fe vce(cluster entity_id)
```

|                                   |                  |   |     |
|-----------------------------------|------------------|---|-----|
| Fixed-effects (within) regression | Number of obs    | = | 198 |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2   |

|                               |                                                          |
|-------------------------------|----------------------------------------------------------|
| R-squared:                    | Obs per group:                                           |
| Within = <b>0.1673</b>        | min = <b>99</b>                                          |
|                               | avg = <b>99.0</b>                                        |
| Overall = <b>0.1107</b>       | max = <b>99</b>                                          |
| corr(u_i, Xb) = <b>0.0000</b> |                                                          |
|                               | $F(1, 1) = .$                                            |
|                               | Prob > F = .                                             |
|                               | (Std. err. adjusted for 2 clusters in <b>entity_id</b> ) |

|                   | Robust      |                                   |        |       |                      |          |
|-------------------|-------------|-----------------------------------|--------|-------|----------------------|----------|
| Delinquency_rate  | Coefficient | std. err.                         | t      | P> t  | [95% conf. interval] |          |
| Unemployment_rate |             |                                   |        |       |                      |          |
| L1.               | .0029104    | .0001406                          | 20.70  | 0.031 | .0011236             | .0046973 |
| L2.               | .000579     | .0000131                          | 44.32  | 0.014 | .000413              | .000745  |
| L3.               | -.0000516   | .0001762                          | -0.29  | 0.819 | -.0022905            | .0021874 |
| L4.               | -.0022701   | .0002148                          | -10.57 | 0.060 | -.0049991            | .0004589 |
| _cons             | .0183929    | .001502                           | 12.25  | 0.052 | -.0006925            | .0374782 |
| sigma_u           | .0117044    |                                   |        |       |                      |          |
| sigma_e           | .01072742   |                                   |        |       |                      |          |
| rho               | .54347107   | (fraction of variance due to u_i) |        |       |                      |          |

```
18 . xtreg Delinquency_rate Unemployment_rate L(1/4).Unemployment_rate, fe vce(cluster entity_id)
```

```

Fixed-effects (within) regression                               Number of obs   =       198
Group variable: entity_id                                   Number of groups =        2

R-squared:                                                    Obs per group:
    Within = 0.2379                                           min =           99
    avg     = 99.0                                           max =          999
    Overall = 0.1574                                           max =           99

corr(u_i, Xb) = 0.0000                                       F(1, 1)         =          .
                                                            Prob > F         =          .

(Std. err. adjusted for 2 clusters in entity_id)

```

|                   | Robust      |                                   |        |       |                      |          |
|-------------------|-------------|-----------------------------------|--------|-------|----------------------|----------|
| Delinquency_rate  | Coefficient | std. err.                         | t      | P> t  | [95% conf. interval] |          |
| Unemployment_rate |             |                                   |        |       |                      |          |
| —.                | .0029807    | .0005411                          | 5.51   | 0.114 | -.0038946            | .0098559 |
| L1.               | .0006722    | .0002653                          | 2.53   | 0.239 | -.002699             | .0040434 |
| L2.               | .0004181    | .0000423                          | 9.88   | 0.064 | -.0001196            | .0009557 |
| L3.               | -.0003017   | .0002221                          | -1.36  | 0.404 | -.0031232            | .0025199 |
| L4.               | -.0021986   | .0002024                          | -10.86 | 0.058 | -.0047698            | .0003726 |
| _cons             | .0160797    | .001086                           | 14.81  | 0.043 | .0022803             | .0298791 |
| sigma_u           | .0117044    |                                   |        |       |                      |          |
| sigma_e           | .01028957   |                                   |        |       |                      |          |
| rho               | .56406319   | (fraction of variance due to u_i) |        |       |                      |          |

```
19 . * unemployment is a lagging indicator so choose lag 2
```

20 . \* should we include real and nominal?

```
21 . xtreg Delinquency_rate L(3).Real_GDP_growth, fe vce(cluster entity_id)
```

```

Fixed-effects (within) regression                               Number of obs   =       200
Group variable: entity_id                                   Number of groups =        2

R-squared:                                                     Obs per group:
    Within = 0.0366                                           min =           100
    Overall = 0.0241                                           avg =          100.0
                                                                max =           100

                                                                F(1, 1)         =       181.96
                                                                Prob > F         =       0.0471

corr(u_i, Xb) = 0.0000

(Std. err. adjusted for 2 clusters in entity_id)

```

|                        | Coefficient | Robust<br>std. err. | t      | P> t  | [95% conf. interval] |           |
|------------------------|-------------|---------------------|--------|-------|----------------------|-----------|
| Real_GDP_growth<br>L3. | -.0004372   | .0000324            | -13.49 | 0.047 | -.000849             | -.0000254 |
| _cons                  | .0261601    | .000074             | 353.56 | 0.002 | .0252199             | .0271002  |

```

sigma_u      .01173656
sigma_e      .01142944
rho          .5132548   (fraction of variance due to u_i)

```

22 . xtreg Delinquency\_rate L(3).Nominal\_GDP\_growth, fe vce(cluster entity\_id)

```

Fixed-effects (within) regression      Number of obs   =      200
Group variable: entity_id              Number of groups =       2

```

```

R-squared:                               Obs per group:
  Within = 0.0570                        min =      100
  Overall = 0.0376                       avg =     100.0
                                         max =      100

```

```

corr(u_i, Xb) = 0.0000                  F(1, 1)         =     841.85
                                         Prob > F         =     0.0219

```

(Std. err. adjusted for 2 clusters in entity\_id)

| Delinquency_rate          | Coefficient | Robust<br>std. err. | t      | P> t  | [95% conf. interval] |                                   |
|---------------------------|-------------|---------------------|--------|-------|----------------------|-----------------------------------|
| Nominal_GDP_growth<br>L3. | -.0004743   | .0000163            | -29.01 | 0.022 | -.0006821            | -.0002666                         |
| _cons                     | .0273786    | .0000764            | 358.38 | 0.002 | .0264079             | .0283493                          |
| sigma_u                   | .01173656   |                     |        |       |                      |                                   |
| sigma_e                   | .01130751   |                     |        |       |                      |                                   |
| rho                       | .51861216   |                     |        |       |                      | (fraction of variance due to u_i) |

23 . xtreg Delinquency\_rate L(3).Real\_GDP\_growth L(3).Nominal\_GDP\_growth, fe vce(cluster entity\_id)

```

Fixed-effects (within) regression      Number of obs   =      200
Group variable: entity_id              Number of groups =       2

```

```

R-squared:                               Obs per group:
  Within = 0.0773                        min =      100
  Overall = 0.0510                       avg =     100.0
                                         max =      100

```

```

corr(u_i, Xb) = 0.0000                  F(1, 1)         =      .
                                         Prob > F         =      .

```

(Std. err. adjusted for 2 clusters in entity\_id)

| Delinquency_rate          | Coefficient | Robust<br>std. err. | t      | P> t  | [95% conf. interval] |                                   |
|---------------------------|-------------|---------------------|--------|-------|----------------------|-----------------------------------|
| Real_GDP_growth<br>L3.    | .0012007    | .000195             | 6.16   | 0.103 | -.0012773            | .0036788                          |
| Nominal_GDP_growth<br>L3. | -.0014786   | .0001467            | -10.08 | 0.063 | -.003343             | .0003857                          |
| _cons                     | .0293303    | .0002404            | 122.00 | 0.005 | .0262755             | .0323851                          |
| sigma_u                   | .01173656   |                     |        |       |                      |                                   |
| sigma_e                   | .01121377   |                     |        |       |                      |                                   |
| rho                       | .52276721   |                     |        |       |                      | (fraction of variance due to u_i) |

24 . \* how about real and nominal affect unemployment?

25 . xtreg Delinquency\_rate L(2).Unemployment\_rate, fe vce(cluster entity\_id)

```

Fixed-effects (within) regression      Number of obs   =      202
Group variable: entity_id              Number of groups =       2

```

```

R-squared:                               Obs per group:
  Within = 0.0409                        min =      101
  Overall = 0.0269                       avg =     101.0
                                         max =      101

```

```

corr(u_i, Xb) = -0.0000                  F(1, 1)         =     28.79
                                         Prob > F         =     0.1173

```

(Std. err. adjusted for 2 clusters in entity\_id)

| Delinquency_rate  | Coefficient | Robust<br>std. err. | t | P> t | [95% conf. interval] |  |
|-------------------|-------------|---------------------|---|------|----------------------|--|
| Unemployment_rate |             |                     |   |      |                      |  |

|         |           |                                   |       |       |           |          |
|---------|-----------|-----------------------------------|-------|-------|-----------|----------|
| L2.     | .0012011  | .0002238                          | 5.37  | 0.117 | -.0016432 | .0040453 |
| _cons   | .0184308  | .0012697                          | 14.52 | 0.044 | .0022977  | .034564  |
| sigma_u | .01178278 |                                   |       |       |           |          |
| sigma_e | .01137922 |                                   |       |       |           |          |
| rho     | .51741805 | (fraction of variance due to u_i) |       |       |           |          |

```
26 . xtreg Delinquency_rate L(3).Real_GDP_growth L(2).Unemployment_rate, fe vce(cluster entity_id)
```

```
Fixed-effects (within) regression      Number of obs   =      200
Group variable: entity_id             Number of groups =       2
```

|                  |                |
|------------------|----------------|
| R-squared:       | Obs per group: |
| Within = 0.0667  | min = 100      |
|                  | avg = 100.0    |
| Overall = 0.0440 | max = 100      |

|                               |                |   |   |
|-------------------------------|----------------|---|---|
|                               | <u>F(1, 1)</u> | = | . |
| corr(u_i, Xb) = <b>0.0000</b> | Prob > F       | = | . |

(Std. err. adjusted for 2 clusters in **entity\_id**)

|                          | Robust      |                                   |       |       |                      |          |
|--------------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Delinquency_rate         | Coefficient | std. err.                         | t     | P> t  | [95% conf. interval] |          |
| Real_GDP_growth<br>L3.   | -.0003523   | .0000518                          | -6.80 | 0.093 | -.0010101            | .0003055 |
| Unemployment_rate<br>L2. | .0010549    | .0002396                          | 4.40  | 0.142 | -.0019891            | .0040989 |
| _cons                    | .0199647    | .0014811                          | 13.48 | 0.047 | .0011457             | .0387838 |
| sigma_u                  | .01173656   |                                   |       |       |                      |          |
| sigma_e                  | .01127791   |                                   |       |       |                      |          |
| rho                      | .51992073   | (fraction of variance due to u_i) |       |       |                      |          |

```
27 . xtreg Delinquency_rate L(3).Nominal_GDP_growth L(2).Unemployment_rate, fe vce(cluster entity_id)
```

```
Fixed-effects (within) regression      Number of obs   =      200
Group variable: entity_id           Number of groups =       2
```

|                         |                    |
|-------------------------|--------------------|
| R-squared:              | Obs per group:     |
| Within = <b>0.0787</b>  | min = <b>100</b>   |
|                         | avg = <b>100.0</b> |
| Overall = <b>0.0519</b> | max = <b>100</b>   |

|                        |          |   |   |
|------------------------|----------|---|---|
| corr(u_i, Xb) = 0.0000 | F(1, 1)  | = | . |
|                        | Prob > F | = | . |

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate          | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|---------------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Nominal_GDP_growth<br>L3. | -.0003865   | .00004                            | -9.66 | 0.066 | -.0008949            | .0001222 |
| Unemployment_rate<br>L2.  | .0009141    | .0002458                          | 3.72  | 0.167 | -.0022096            | .0040377 |
| _cons                     | .0217678    | .0015856                          | 13.73 | 0.046 | .0016212             | .0419143 |
| sigma_u                   | .01173656   |                                   |       |       |                      |          |
| sigma_e                   | .01120508   |                                   |       |       |                      |          |
| rho                       | .52315431   | (fraction of variance due to u_i) |       |       |                      |          |

28 . \* seems like nominal gives more impact than real GDP

29 . \* consider inflation to partition nominal = real + inflation

```
30 . xtreg Delinquency_rate CPI_inflation_rate, fe vce(cluster entity_id)
```

```
Fixed-effects (within) regression      Number of obs   =      206
Group variable: entity_id             Number of groups =       2
```

|                         |                    |
|-------------------------|--------------------|
| R-squared:              | Obs per group:     |
| Within = <b>0.0276</b>  | min = <b>103</b>   |
|                         | avg = <b>103.0</b> |
| Overall = <b>0.0180</b> | max = <b>103</b>   |

corr(u\_i, Xb) = **0.0000**

F(1, 1) = **18.49**  
Prob > F = **0.1454**

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate   | Coefficient      | Robust<br>std. err.               | t            | P> t         | [95% conf. interval] |                 |
|--------------------|------------------|-----------------------------------|--------------|--------------|----------------------|-----------------|
| CPI_inflation_rate | <b>-.0007762</b> | <b>.0001805</b>                   | <b>-4.30</b> | <b>0.145</b> | <b>-.0030696</b>     | <b>.0015172</b> |
| _cons              | <b>.0273888</b>  | <b>.0004679</b>                   | <b>58.54</b> | <b>0.011</b> | <b>.0214439</b>      | <b>.0333338</b> |
| sigma_u            | <b>.01188763</b> |                                   |              |              |                      |                 |
| sigma_e            | <b>.01139595</b> |                                   |              |              |                      |                 |
| rho                | <b>.52110768</b> | (fraction of variance due to u_i) |              |              |                      |                 |

31 . xtreg Delinquency\_rate L(1/4).CPI\_inflation\_rate, fe vce(cluster entity\_id)

Fixed-effects (within) regression

Group variable: **entity\_id**

R-squared:

Within = **0.0562**

Overall = **0.0372**

Number of obs = **198**  
Number of groups = **2**  
Obs per group: min = **99**, avg = **99.0**, max = **99**

F(1, 1) = **.**  
Prob > F = **.**

corr(u\_i, Xb) = **0.0000**

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate   | Coefficient      | Robust<br>std. err.               | t             | P> t         | [95% conf. interval] |                  |
|--------------------|------------------|-----------------------------------|---------------|--------------|----------------------|------------------|
| CPI_inflation_rate |                  |                                   |               |              |                      |                  |
| L1.                | <b>-.0006531</b> | <b>.0001742</b>                   | <b>-3.75</b>  | <b>0.166</b> | <b>-.002867</b>      | <b>.0015608</b>  |
| L2.                | <b>-.0004239</b> | <b>.0000215</b>                   | <b>-19.74</b> | <b>0.032</b> | <b>-.0006967</b>     | <b>-.0001511</b> |
| L3.                | <b>-.0003742</b> | <b>.0001272</b>                   | <b>-2.94</b>  | <b>0.209</b> | <b>-.0019906</b>     | <b>.0012421</b>  |
| L4.                | <b>-.0000205</b> | <b>.0002205</b>                   | <b>-0.09</b>  | <b>0.941</b> | <b>-.0028219</b>     | <b>.002781</b>   |
| _cons              | <b>.0288462</b>  | <b>.0005061</b>                   | <b>57.00</b>  | <b>0.011</b> | <b>.0224159</b>      | <b>.0352766</b>  |
| sigma_u            | <b>.0117044</b>  |                                   |               |              |                      |                  |
| sigma_e            | <b>.0114208</b>  |                                   |               |              |                      |                  |
| rho                | <b>.51226201</b> | (fraction of variance due to u_i) |               |              |                      |                  |

32 . xtreg Delinquency\_rate CPI\_inflation\_rate L(1/4).CPI\_inflation\_rate, fe vce(cluster entity\_id)

Fixed-effects (within) regression

Group variable: **entity\_id**

R-squared:

Within = **0.0655**

Overall = **0.0433**

Number of obs = **198**  
Number of groups = **2**  
Obs per group: min = **99**, avg = **99.0**, max = **99**

F(1, 1) = **.**  
Prob > F = **.**

corr(u\_i, Xb) = **-0.0000**

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate   | Coefficient      | Robust<br>std. err.               | t             | P> t         | [95% conf. interval] |                  |
|--------------------|------------------|-----------------------------------|---------------|--------------|----------------------|------------------|
| CPI_inflation_rate |                  |                                   |               |              |                      |                  |
| _.                 | <b>-.0004938</b> | <b>.0002145</b>                   | <b>-2.30</b>  | <b>0.261</b> | <b>-.0032189</b>     | <b>.0022314</b>  |
| L1.                | <b>-.0004594</b> | <b>.0000906</b>                   | <b>-5.07</b>  | <b>0.124</b> | <b>-.0016099</b>     | <b>.0006912</b>  |
| L2.                | <b>-.0004299</b> | <b>.0000189</b>                   | <b>-22.71</b> | <b>0.028</b> | <b>-.0006704</b>     | <b>-.0001893</b> |
| L3.                | <b>-.0003021</b> | <b>.0001588</b>                   | <b>-1.90</b>  | <b>0.308</b> | <b>-.0023205</b>     | <b>.0017162</b>  |
| L4.                | <b>-.000053</b>  | <b>.0002069</b>                   | <b>-0.26</b>  | <b>0.840</b> | <b>-.002682</b>      | <b>.0025759</b>  |
| _cons              | <b>.0295282</b>  | <b>.0002112</b>                   | <b>139.83</b> | <b>0.005</b> | <b>.026845</b>       | <b>.0322113</b>  |
| sigma_u            | <b>.0117044</b>  |                                   |               |              |                      |                  |
| sigma_e            | <b>.01139406</b> |                                   |               |              |                      |                  |
| rho                | <b>.51343324</b> | (fraction of variance due to u_i) |               |              |                      |                  |

33 .

| Deliquency_rate             | Coefficient          | std. err.                         | t              | P> t           | [95% conf. interval]  |                      |
|-----------------------------|----------------------|-----------------------------------|----------------|----------------|-----------------------|----------------------|
| Real_GDP_growth<br>L3.      | -.0003764            | .000015                           | -25.15         | 0.025          | -.0005665             | -.0001862            |
| CPI_inflation_rate<br>_cons | -.0006774<br>.027756 | .0001952<br>.0005341              | -3.47<br>51.97 | 0.179<br>0.012 | -.0031577<br>.0209697 | .0018029<br>.0345423 |
| sigma_u                     | .01173656            |                                   |                |                |                       |                      |
| sigma_e                     | .01133605            |                                   |                |                |                       |                      |
| rho                         | .51735353            | (fraction of variance due to u_i) |                |                |                       |                      |

1 . xtreg Deliquency\_rate L(3).Real\_GDP\_growth L(1/4).CPI\_inflation\_rate, fe vce(cluster entity\_id)

Fixed-effects (within) regression                      Number of obs       =        198  
Group variable: **entity\_id**                      Number of groups    =        2

R-squared:                                              Obs per group:                      min =        99  
Within   = 0.0801                                              avg   =       99.0  
Overall   = 0.0530                                              max   =       99

corr(u\_i, Xb) = 0.0000                                              F(1, 1)                      =        .  
Prob > F                                              =        .

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Deliquency_rate           | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|---------------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Real_GDP_growth<br>L3.    | -.0003904   | .0000424                          | -9.21  | 0.069 | -.000929             | .0001481  |
| CPI_inflation_rate<br>L1. | -.0005848   | .0001673                          | -3.50  | 0.177 | -.0027103            | .0015407  |
| L2.                       | -.0004599   | .0000176                          | -26.11 | 0.024 | -.0006837            | -.0002361 |
| L3.                       | -.0000114   | .0001669                          | -0.07  | 0.957 | -.0021324            | .0021096  |
| L4.                       | -.0002556   | .0001955                          | -1.31  | 0.416 | -.00274              | .0022288  |
| _cons                     | .0293332    | .0004545                          | 64.54  | 0.010 | .0235578             | .0351085  |
| sigma_u                   | .0117044    |                                   |        |       |                      |           |
| sigma_e                   | .01130476   |                                   |        |       |                      |           |
| rho                       | .51736344   | (fraction of variance due to u_i) |        |       |                      |           |

2 . xtreg Deliquency\_rate L(3).Real\_GDP\_growth CPI\_inflation\_rate L(1/4).CPI\_inflation\_rate, fe vce(cluster entity\_id)

Fixed-effects (within) regression                      Number of obs       =        198  
Group variable: **entity\_id**                      Number of groups    =        2

R-squared:                                              Obs per group:                      min =        99  
Within   = 0.0865                                              avg   =       99.0  
Overall   = 0.0572                                              max   =       99

corr(u\_i, Xb) = 0.0000                                              F(1, 1)                      =        .  
Prob > F                                              =        .

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Deliquency_rate          | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|--------------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Real_GDP_growth<br>L3.   | -.000368    | .0000312                          | -11.80 | 0.054 | -.0007644            | .0000283  |
| CPI_inflation_rate<br>_. | -.0004116   | .0002081                          | -1.98  | 0.298 | -.0030555            | .0022322  |
| L1.                      | -.0004272   | .0000881                          | -4.85  | 0.129 | -.0015462            | .0006918  |
| L2.                      | -.0004628   | .0000162                          | -28.59 | 0.022 | -.0006685            | -.0002572 |
| L3.                      | .0000279    | .0001872                          | 0.15   | 0.906 | -.0023511            | .0024069  |
| L4.                      | -.0002693   | .0001891                          | -1.42  | 0.390 | -.0026722            | .0021337  |
| _cons                    | .0298738    | .0001824                          | 163.76 | 0.004 | .0275559             | .0321918  |
| sigma_u                  | .0117044    |                                   |        |       |                      |           |
| sigma_e                  | .01129504   |                                   |        |       |                      |           |
| rho                      | .51779327   | (fraction of variance due to u_i) |        |       |                      |           |

3 . xtreg Deliquency\_rate L(3).Real\_GDP\_growth L(2).CPI\_inflation\_rate, fe vce(cluster entity\_id)

Fixed-effects (within) regression                      Number of obs       =        200



```

Group variable: entity_id                Number of groups =      2

R-squared:                                Obs per group:
  Within = 0.0616                          min =      100
                                           avg =     100.0
  Overall = 0.0406                          max =     100

                                           F(1, 1)      =      .
corr(u_i, Xb) = 0.0000                     Prob > F      =      .

```

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate          | Coefficient      | Robust<br>std. err.               | t             | P> t         | [95% conf. interval] |                  |
|---------------------------|------------------|-----------------------------------|---------------|--------------|----------------------|------------------|
| Real_GDP_growth<br>L3.    | <b>-.0003945</b> | <b>.0000357</b>                   | <b>-11.04</b> | <b>0.058</b> | <b>-.0008487</b>     | <b>.0000596</b>  |
| CPI_inflation_rate<br>L2. | <b>-.0007406</b> | <b>.0000565</b>                   | <b>-13.11</b> | <b>0.048</b> | <b>-.0014586</b>     | <b>-.0000227</b> |
| _cons                     | <b>.0279758</b>  | <b>.0000643</b>                   | <b>434.79</b> | <b>0.001</b> | <b>.0271583</b>      | <b>.0287934</b>  |
| sigma_u                   | <b>.01173656</b> |                                   |               |              |                      |                  |
| sigma_e                   | <b>.01130884</b> |                                   |               |              |                      |                  |
| rho                       | <b>.51855353</b> | (fraction of variance due to u_i) |               |              |                      |                  |

4 . \* test nominal and inflation

5 . xtreg Delinquency\_rate L(3).Nominal\_GDP\_growth L(2).CPI\_inflation\_rate, fe vce(cluster entity\_id)

```

Fixed-effects (within) regression          Number of obs   =     200
Group variable: entity_id                Number of groups =      2

R-squared:                                Obs per group:
  Within = 0.0731                          min =      100
                                           avg =     100.0
  Overall = 0.0482                          max =     100

```

```

                                           F(1, 1)      =      .
corr(u_i, Xb) = -0.0000                     Prob > F      =      .

```

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate          | Coefficient      | Robust<br>std. err.               | t             | P> t         | [95% conf. interval] |                 |
|---------------------------|------------------|-----------------------------------|---------------|--------------|----------------------|-----------------|
| Nominal_GDP_growth<br>L3. | <b>-.0004133</b> | <b>.0000225</b>                   | <b>-18.41</b> | <b>0.035</b> | <b>-.0006987</b>     | <b>-.000128</b> |
| CPI_inflation_rate<br>L2. | <b>-.0006069</b> | <b>.0000604</b>                   | <b>-10.06</b> | <b>0.063</b> | <b>-.0013739</b>     | <b>.00016</b>   |
| _cons                     | <b>.0286612</b>  | <b>.000051</b>                    | <b>562.31</b> | <b>0.001</b> | <b>.0280136</b>      | <b>.0293088</b> |
| sigma_u                   | <b>.01173656</b> |                                   |               |              |                      |                 |
| sigma_e                   | <b>.0112392</b>  |                                   |               |              |                      |                 |
| rho                       | <b>.52163673</b> | (fraction of variance due to u_i) |               |              |                      |                 |

6 . \* although R2 fits better, nominal includes inflation. So, it is overfitting. Then choose real + inflation is better

7 . \* compare again with nominal GDP, real+GDP give better fits (higher R2)

8 . xtreg Delinquency\_rate L(3).Nominal\_GDP\_growth, fe vce(cluster entity\_id)

```

Fixed-effects (within) regression          Number of obs   =     200
Group variable: entity_id                Number of groups =      2

R-squared:                                Obs per group:
  Within = 0.0570                          min =      100
                                           avg =     100.0
  Overall = 0.0376                          max =     100

```

```

                                           F(1, 1)      =    841.85
corr(u_i, Xb) = 0.0000                     Prob > F      =    0.0219

```

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate          | Coefficient      | Robust<br>std. err. | t             | P> t         | [95% conf. interval] |                  |
|---------------------------|------------------|---------------------|---------------|--------------|----------------------|------------------|
| Nominal_GDP_growth<br>L3. | <b>-.0004743</b> | <b>.0000163</b>     | <b>-29.01</b> | <b>0.022</b> | <b>-.0006821</b>     | <b>-.0002666</b> |

|         |           |                                   |        |       |          |          |
|---------|-----------|-----------------------------------|--------|-------|----------|----------|
| _cons   | .0273786  | .0000764                          | 358.38 | 0.002 | .0264079 | .0283493 |
| sigma_u | .01173656 |                                   |        |       |          |          |
| sigma_e | .01130751 |                                   |        |       |          |          |
| rho     | .51861216 | (fraction of variance due to u_i) |        |       |          |          |

9 . \* combine real, CPI and Unemployment

10 . xtreg Delinquency\_rate L(3).Real\_GDP\_growth L(2).CPI\_inflation\_rate L(2).Unemployment\_rate, fe vce(cluster entity\_id)

Fixed-effects (within) regression                      Number of obs       =       200  
Group variable: **entity\_id**                      Number of groups    =       2

R-squared:                                              Obs per group:                      min =       100  
Within   = 0.0799                                              avg   =      100.0  
Overall = 0.0527                                              max   =      100

corr(u\_i, Xb) = -0.0000                       $\frac{F(1, 1)}{\text{Prob} > F}$                       =       .

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate          | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|---------------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Real_GDP_growth<br>L3.    | -.0003362   | .0000521                          | -6.46  | 0.098 | -.0009979            | .0003255  |
| CPI_inflation_rate<br>L2. | -.000559    | 6.07e-06                          | -92.08 | 0.007 | -.0006361            | -.0004818 |
| Unemployment_rate<br>L2.  | .0008549    | .000238                           | 3.59   | 0.173 | -.0021692            | .0038791  |
| _cons                     | .0225096    | .0014572                          | 15.45  | 0.041 | .0039938             | .0410254  |
| sigma_u                   | .01173656   |                                   |        |       |                      |           |
| sigma_e                   | .01122677   |                                   |        |       |                      |           |
| rho                       | .52218921   | (fraction of variance due to u_i) |        |       |                      |           |

11 . xtreg Delinquency\_rate L(3).Nominal\_GDP\_growth L(2).CPI\_inflation\_rate L(2).Unemployment\_rate, fe vce(cluster entity\_id)

Fixed-effects (within) regression                      Number of obs       =       200  
Group variable: **entity\_id**                      Number of groups    =       2

R-squared:                                              Obs per group:                      min =       100  
Within   = 0.0876                                              avg   =      100.0  
Overall = 0.0578                                              max   =      100

corr(u\_i, Xb) = -0.0000                       $\frac{F(1, 1)}{\text{Prob} > F}$                       =       .

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate          | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|---------------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Nominal_GDP_growth<br>L3. | -.0003538   | .0000412                          | -8.58  | 0.074 | -.0008775            | .0001699  |
| CPI_inflation_rate<br>L2. | -.0004638   | .0000155                          | -29.83 | 0.021 | -.0006614            | -.0002662 |
| Unemployment_rate<br>L2.  | .000769     | .0002416                          | 3.18   | 0.194 | -.0023008            | .0038388  |
| _cons                     | .0236386    | .0015269                          | 15.48  | 0.041 | .0042377             | .0430395  |
| sigma_u                   | .01173656   |                                   |        |       |                      |           |
| sigma_e                   | .0111797    |                                   |        |       |                      |           |
| rho                       | .5242855    | (fraction of variance due to u_i) |        |       |                      |           |

12 . \* test dependency of delinquency rate to BBB

13 . xtreg Delinquency\_rate BBB\_corporate\_yield, fe vce(cluster entity\_id)

Fixed-effects (within) regression                      Number of obs       =       206  
Group variable: **entity\_id**                      Number of groups    =       2

| (Std. err. adjusted for 2 clusters in <b>entity_id</b> ) |                  |                                   |              |              |                      |                 |
|----------------------------------------------------------|------------------|-----------------------------------|--------------|--------------|----------------------|-----------------|
| Delinquency_rate                                         | Coefficient      | Robust<br>std. err.               | t            | P> t         | [95% conf. interval] |                 |
| BBB_corporate_yield                                      | <b>.0050081</b>  | <b>.0014336</b>                   | <b>3.49</b>  | <b>0.177</b> | <b>-.0132072</b>     | <b>.0232233</b> |
| _cons                                                    | <b>-.0008159</b> | <b>.0074977</b>                   | <b>-0.11</b> | <b>0.931</b> | <b>-.0960833</b>     | <b>.0944515</b> |
| sigma_u                                                  | <b>.01188763</b> |                                   |              |              |                      |                 |
| sigma_e                                                  | <b>.00858087</b> |                                   |              |              |                      |                 |
| rho                                                      | <b>.65744445</b> | (fraction of variance due to u_i) |              |              |                      |                 |

(Std. err. adjusted for 2 clusters in **entity\_id**)

|                     |             | Robust                            |       |       |                      |          |
|---------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Delinquency_rate    | Coefficient | std. err.                         | t     | P> t  | [95% conf. interval] |          |
| BBB_corporate_yield |             |                                   |       |       |                      |          |
| L1.                 | .0012156    | .0020133                          | 0.60  | 0.654 | -.0243663            | .0267976 |
| L2.                 | .0013657    | .0002592                          | 5.27  | 0.119 | -.0019281            | .0046596 |
| L3.                 | .0003248    | .0001197                          | 2.71  | 0.225 | -.0011963            | .0018459 |
| L4.                 | .0037727    | .0003255                          | 11.59 | 0.055 | -.0003627            | .0079081 |
| _cons               | -.0095942   | .0066797                          | -1.44 | 0.387 | -.0944677            | .0752793 |
| sigma_u             | .0117044    |                                   |       |       |                      |          |
| sigma_e             | .00630185   |                                   |       |       |                      |          |
| rho                 | .77525811   | (fraction of variance due to u_i) |       |       |                      |          |

(Std. err. adjusted for 2 clusters in **entity\_id**)

|                     |             | Robust                            |       |       |                      |          |
|---------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Delinquency_rate    | Coefficient | std. err.                         | t     | P> t  | [95% conf. interval] |          |
| BBB_corporate_yield |             |                                   |       |       |                      |          |
| —.                  | -.0005433   | .0015485                          | -0.35 | 0.785 | -.0202192            | .0191327 |
| L1.                 | .0018476    | .0002171                          | 8.51  | 0.074 | -.0009104            | .0046057 |
| L2.                 | .0012615    | .0000372                          | 33.93 | 0.019 | .0007891             | .001734  |
| L3.                 | .0002351    | .0001356                          | 1.73  | 0.333 | -.0014873            | .0019575 |
| L4.                 | .0038324    | .0004963                          | 7.72  | 0.082 | -.0024741            | .0101389 |
| _cons               | -.0093729   | .0073278                          | -1.28 | 0.422 | -.1024818            | .0837361 |
| sigma_u             | .0117044    |                                   |       |       |                      |          |
| sigma_e             | .00631297   |                                   |       |       |                      |          |
| rho                 | .77464298   | (fraction of variance due to u_i) |       |       |                      |          |

```
16 . * test dependency of delinquency rate to Mortgage rate
```

|                                   |                  |               |                |              |      |               |
|-----------------------------------|------------------|---------------|----------------|--------------|------|---------------|
| Fixed-effects (within) regression | Number of obs    | =             | 206            |              |      |               |
| Group variable: <b>entity_id</b>  | Number of groups | =             | 2              |              |      |               |
| R-squared:                        |                  |               |                |              |      |               |
| Within                            | =                | <b>0.1894</b> | Obs per group: |              |      |               |
|                                   |                  |               | min =          | <b>103</b>   |      |               |
|                                   |                  |               | avg =          | <b>103.0</b> |      |               |
| Overall                           | =                | <b>0.1232</b> | max =          | <b>103</b>   |      |               |
| F(1, 1)                           |                  |               |                | =            | 7.35 |               |
| corr(u_i, Xb) = <b>0.0000</b>     | Prob > F         |               |                |              | =    | <b>0.2250</b> |

|                | Coefficient      | Robust<br>std. err.               | t           | P> t         | [95% conf. interval]             |
|----------------|------------------|-----------------------------------|-------------|--------------|----------------------------------|
| Delinquency_~e |                  |                                   |             |              |                                  |
| Mortgage_rate  | <b>.0036036</b>  | <b>.0013295</b>                   | <b>2.71</b> | <b>0.225</b> | <b>-.0132889</b> <b>.0204966</b> |
| _cons          | <b>.0066452</b>  | <b>.0069106</b>                   | <b>0.96</b> | <b>0.512</b> | <b>-.0811627</b> <b>.0944531</b> |
| sigma_u        | <b>.01188763</b> |                                   |             |              |                                  |
| sigma_e        | <b>.01040488</b> |                                   |             |              |                                  |
| rho            | <b>.56622031</b> | (fraction of variance due to u_i) |             |              |                                  |

|                                   |                  |   |      |
|-----------------------------------|------------------|---|------|
| Fixed-effects (within) regression | Number of obs    | = | 198  |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2    |
| R-squared:                        | Obs per group:   |   |      |
| Within = <b>0.4268</b>            | min =            |   | 99   |
|                                   | avg =            |   | 99.0 |
| Overall = <b>0.2824</b>           | max =            |   | 99   |
|                                   | $F(1, 1)$        | = | .    |
| corr(u_i, Xb) = <b>-0.0000</b>    | Prob > F         | = |      |

|               | Coefficient | Robust<br>std. err. | t     | P> t  | [95% conf. interval] |                                   |
|---------------|-------------|---------------------|-------|-------|----------------------|-----------------------------------|
| Mortgage_rate |             |                     |       |       |                      |                                   |
| L1.           | -.0028628   | .000406             | -7.05 | 0.090 | -.0080221            | .0022965                          |
| L2.           | .0009352    | .0009605            | 0.97  | 0.508 | -.0112693            | .0131397                          |
| L3.           | -.0008112   | .0005259            | -1.54 | 0.366 | -.0074932            | .0058707                          |
| L4.           | .007639     | .0004313            | 17.71 | 0.036 | .0021584             | .0131195                          |
| _cons         | -.0002145   | .0074482            | -0.03 | 0.982 | -.0948523            | .0944233                          |
| sigma_u       | .0117044    |                     |       |       |                      |                                   |
| sigma_e       | .0089052    |                     |       |       |                      |                                   |
| rho           | .6336049    |                     |       |       |                      | (fraction of variance due to u_i) |

|                                   |                  |   |             |
|-----------------------------------|------------------|---|-------------|
| Fixed-effects (within) regression | Number of obs    | = | 198         |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2           |
| R-squared:                        | Obs per group:   |   |             |
| Within = <b>0.4305</b>            | min =            |   | <b>99</b>   |
|                                   | avg =            |   | <b>99.0</b> |
| Overall = <b>0.2849</b>           | max =            |   | <b>99</b>   |
|                                   | <u>F(1, 1)</u>   | = | .           |
| corr(u_i, Xb) = <b>-0.0000</b>    | Prob > F         | = | .           |

|               | Coefficient | Robust<br>std. err. | t     | P> t  | [95% conf. interval] |
|---------------|-------------|---------------------|-------|-------|----------------------|
| Mortgage_rate |             |                     |       |       |                      |
| _             | -.002162    | .0003847            | -5.62 | 0.112 | -.0070502 .0027262   |
| L1.           | -.0003782   | .0008492            | -0.45 | 0.733 | -.0111687 .0104124   |
| L2.           | .0005746    | .0008989            | 0.64  | 0.638 | -.0108464 .0119957   |
| L3.           | -.0006612   | .0005539            | -1.19 | 0.444 | -.0076997 .0063774   |
| L4.           | .0074067    | .0004738            | 15.63 | 0.041 | .0013866 .0134267    |
| _cons         | .0003862    | .0073606            | 0.05  | 0.967 | -.0931397 .0939121   |

|         |           |                                   |
|---------|-----------|-----------------------------------|
| sigma_u | .0117044  |                                   |
| sigma_e | .00889464 |                                   |
| rho     | .63391124 | (fraction of variance due to u_i) |

20 . \* test dependency of delinquency rate to Prime rate

21 . xtreg Delinquency\_rate Prime\_rate, fe vce(cluster entity\_id)

|                                   |                  |   |               |
|-----------------------------------|------------------|---|---------------|
| Fixed-effects (within) regression | Number of obs    | = | 206           |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2             |
| R-squared:                        | Obs per group:   |   |               |
| Within = <b>0.0020</b>            | min =            |   | <b>103</b>    |
|                                   | avg =            |   | <b>103.0</b>  |
| Overall = <b>0.0013</b>           | max =            |   | <b>103</b>    |
|                                   | F(1, 1)          | = | <b>0.07</b>   |
| corr(u_i, Xb) = <b>0.0000</b>     | Prob > F         | = | <b>0.8337</b> |

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency~e | Coefficient | Robust<br>std. err. | t    | P> t  | [95% conf. interval] |                                   |
|---------------|-------------|---------------------|------|-------|----------------------|-----------------------------------|
| Prime_rate    | .0002605    | .0009743            | 0.27 | 0.834 | -.0121193            | .0126404                          |
| _cons         | .0240474    | .0049709            | 4.84 | 0.130 | -.0391139            | .0872087                          |
| sigma_u       | .01188763   |                     |      |       |                      |                                   |
| sigma_e       | .01154469   |                     |      |       |                      |                                   |
| rho           | .51463241   |                     |      |       |                      | (fraction of variance due to u_i) |

22 . xtreg Delinquency\_rate L(1/4).Prime\_rate, fe vce(cluster entity\_id)

|                                   |                  |   |             |
|-----------------------------------|------------------|---|-------------|
| Fixed-effects (within) regression | Number of obs    | = | 198         |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2           |
| R-squared:                        | Obs per group:   |   |             |
| Within = <b>0.2464</b>            | min =            |   | <b>99</b>   |
|                                   | avg =            |   | <b>99.0</b> |
| Overall = <b>0.1631</b>           | max =            |   | <b>99</b>   |
|                                   | F(1, 1)          | = | .           |
| corr(u_i, Xb) = <b>0.0000</b>     | Prob > F         | = | .           |

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency~e | Coefficient | Robust<br>std. err. | t      | P> t  | [95% conf. interval] |                                   |
|---------------|-------------|---------------------|--------|-------|----------------------|-----------------------------------|
| Prime_rate    |             |                     |        |       |                      |                                   |
| L1.           | -.0019911   | .0001239            | -16.07 | 0.040 | -.0035651            | -.000417                          |
| L2.           | .0019256    | .0014153            | 1.36   | 0.404 | -.0160578            | .0199091                          |
| L3.           | -.0082391   | .0005418            | -15.21 | 0.042 | -.0151228            | -.0013554                         |
| L4.           | .0097621    | .0000614            | 158.91 | 0.004 | .0089816             | .0105427                          |
| _cons         | .0176623    | .0052586            | 3.36   | 0.184 | -.049154             | .0844787                          |
| sigma_u       | .0117044    |                     |        |       |                      |                                   |
| sigma_e       | .01020492   |                     |        |       |                      |                                   |
| rho           | .56812117   |                     |        |       |                      | (fraction of variance due to u_i) |

23 . xtreg Delinquency\_rate Prime\_rate L(1/4).Prime\_rate, fe vce(cluster entity\_id)

|                                   |                  |   |             |
|-----------------------------------|------------------|---|-------------|
| Fixed-effects (within) regression | Number of obs    | = | 198         |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2           |
| R-squared:                        | Obs per group:   |   |             |
| Within = <b>0.2481</b>            | min =            |   | <b>99</b>   |
|                                   | avg =            |   | <b>99.0</b> |
| Overall = <b>0.1642</b>           | max =            |   | <b>99</b>   |
|                                   | F(1, 1)          | = | .           |
| corr(u_i, Xb) = <b>0.0000</b>     | Prob > F         | = | .           |

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency~e | Coefficient | Robust<br>std. err. | t     | P> t  | [95% conf. interval] |          |
|---------------|-------------|---------------------|-------|-------|----------------------|----------|
| Prime_rate    |             |                     |       |       |                      |          |
| --.           | -.0016126   | .0002922            | -5.52 | 0.114 | -.0053249            | .0020998 |
| L1.           | .0006866    | .0006093            | 1.13  | 0.462 | -.0070558            | .008429  |

|         |           |                                   |        |       |           |           |
|---------|-----------|-----------------------------------|--------|-------|-----------|-----------|
| L2.     | .0007037  | .0011976                          | 0.59   | 0.662 | -.0145135 | .0159208  |
| L3.     | -.0080394 | .000507                           | -15.86 | 0.040 | -.0144812 | -.0015975 |
| L4.     | .0096181  | .0000355                          | 270.93 | 0.002 | .0091671  | .0100692  |
| _cons   | .0181551  | .005183                           | 3.50   | 0.177 | -.0477006 | .0840108  |
| sigma_u | .0117044  |                                   |        |       |           |           |
| sigma_e | .01022052 |                                   |        |       |           |           |
| rho     | .56737164 | (fraction of variance due to u_i) |        |       |           |           |

24 . \* test dependency of delinquency rate to 3 month Treasury

```
25 . xtreg Delinquency_rate _3_month_Treasury_rate, fe vce(cluster entity_id)
```

```
Fixed-effects (within) regression      Number of obs   =      206
Group variable: entity_id             Number of groups =       2
```

|                  |                |
|------------------|----------------|
| R-squared:       | Obs per group: |
| Within = 0.0010  | min = 103      |
|                  | avg = 103.0    |
| Overall = 0.0006 | max = 103      |

|                         |          |   |        |
|-------------------------|----------|---|--------|
|                         | F(1, 1)  | = | 0.04   |
| corr(u_i, Xb) = -0.0000 | Prob > F | = | 0.8726 |

(Std. err. adjusted for 2 clusters in **entity\_id**)

|                        | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|------------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Delinquency_rate       |             |                                   |       |       |                      |          |
| _3_month_Treasury_rate | .0001879    | .0009261                          | 0.20  | 0.873 | -.0115795            | .0119553 |
| _cons                  | .0250258    | .0017299                          | 14.47 | 0.044 | .0030447             | .0470068 |
| sigma_u                | .01188763   |                                   |       |       |                      |          |
| sigma_e                | .01155074   |                                   |       |       |                      |          |
| rho                    | .51437072   | (fraction of variance due to u_i) |       |       |                      |          |

```
26 . xtreg Delinquency_rate L(1/4)._3_month_Treasury_rate, fe vce(cluster entity_id)
```

|                                   |                  |   |     |
|-----------------------------------|------------------|---|-----|
| Fixed-effects (within) regression | Number of obs    | = | 198 |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2   |

|                         |  |                |             |
|-------------------------|--|----------------|-------------|
| R-squared:              |  | Obs per group: |             |
| Within = <b>0.2076</b>  |  | min =          | <b>99</b>   |
|                         |  | avg =          | <b>99.0</b> |
| Overall = <b>0.1374</b> |  | max =          | <b>99</b>   |

|                         |          |   |   |
|-------------------------|----------|---|---|
|                         | F(1, 1)  | = | . |
| corr(u_i, Xb) = -0.0000 | Prob > F | = | . |

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate       | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |          |
|------------------------|-------------|-----------------------------------|--------|-------|----------------------|----------|
| _3_month_Treasury_rate |             |                                   |        |       |                      |          |
| L1.                    | -.0022464   | .0002236                          | -10.05 | 0.063 | -.0050878            | .0005951 |
| L2.                    | .0012516    | .0011317                          | 1.11   | 0.468 | -.0131275            | .0156307 |
| L3.                    | -.0063551   | .0005295                          | -12.00 | 0.053 | -.0130836            | .0003734 |
| L4.                    | .0085585    | .0004301                          | 19.90  | 0.032 | .0030939             | .0140231 |
| _cons                  | .02279      | .0017604                          | 12.95  | 0.049 | .0004224             | .0451577 |
| sigma_u                | .0117044    |                                   |        |       |                      |          |
| sigma_e                | .01046467   |                                   |        |       |                      |          |
| rho                    | .55574722   | (fraction of variance due to u_i) |        |       |                      |          |

```
27 . xtreg Delinquency_rate _3_month_Treasury_rate L(1/4)._3_month_Treasury_rate, fe vce(cluster entity_id)
```

```
Fixed-effects (within) regression      Number of obs   =      198
Group variable: entity_id           Number of groups =       2
```

|                         |                   |
|-------------------------|-------------------|
| R-squared:              | Obs per group:    |
| Within = <b>0.2078</b>  | min = <b>99</b>   |
|                         | avg = <b>99.0</b> |
| Overall = <b>0.1375</b> | max = <b>99</b>   |

|                        |                                   |   |   |
|------------------------|-----------------------------------|---|---|
| corr(u_i, Xb) = 0.0000 | $\frac{F(1, 1)}{\text{Prob} > F}$ | = | . |
|                        |                                   | = | . |

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate       | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|------------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| _3_month_Treasury_rate |             |                                   |       |       |                      |          |
| ---                    | -.0005365   | .0005476                          | -0.98 | 0.506 | -.0074942            | .0064211 |
| L1.                    | -.0014244   | .0006147                          | -2.32 | 0.259 | -.0092354            | .0063867 |
| L2.                    | .0009325    | .0008089                          | 1.15  | 0.455 | -.0093461            | .0112112 |
| L3.                    | -.0062591   | .0006289                          | -9.95 | 0.064 | -.0142501            | .0017319 |
| L4.                    | .0084576    | .0005342                          | 15.83 | 0.040 | .0016701             | .015245  |
| _cons                  | .0228534    | .0017003                          | 13.44 | 0.047 | .001249              | .0444578 |
| sigma_u                | .0117044    |                                   |       |       |                      |          |
| sigma_e                | .01049079   |                                   |       |       |                      |          |
| rho                    | .55451588   | (fraction of variance due to u_i) |       |       |                      |          |

28 . \* test dependency of delinquency rate to 5 year Treasury

29 . xtreg Delinquency\_rate \_5\_year\_Treasury\_yield, fe vce(cluster entity\_id)

|                                                          |                  |   |               |
|----------------------------------------------------------|------------------|---|---------------|
| Fixed-effects (within) regression                        | Number of obs    | = | 206           |
| Group variable: <b>entity_id</b>                         | Number of groups | = | 2             |
| R-squared:                                               | Obs per group:   |   |               |
| Within = <b>0.1074</b>                                   | min =            |   | <b>103</b>    |
| Overall = <b>0.0699</b>                                  | avg =            |   | <b>103.0</b>  |
|                                                          | max =            |   | <b>103</b>    |
|                                                          | F(1, 1)          | = | <b>4.49</b>   |
| corr(u_i, Xb) = <b>0.0000</b>                            | Prob > F         | = | <b>0.2807</b> |
| (Std. err. adjusted for 2 clusters in <b>entity_id</b> ) |                  |   |               |

| Delinquency_rate       | Coefficient | Robust<br>std. err.               | t    | P> t  | [95% conf. interval] |          |
|------------------------|-------------|-----------------------------------|------|-------|----------------------|----------|
| _5_year_Treasury_yield | .0025377    | .0011975                          | 2.12 | 0.281 | -.0126774            | .0177527 |
| _cons                  | .0183082    | .0033354                          | 5.49 | 0.115 | -.0240724            | .0606887 |
| sigma_u                | .01188763   |                                   |      |       |                      |          |
| sigma_e                | .01091852   |                                   |      |       |                      |          |
| rho                    | .54241683   | (fraction of variance due to u_i) |      |       |                      |          |

30 . xtreg Delinquency\_rate L(1/4).\_5\_year\_Treasury\_yield, fe vce(cluster entity\_id)

|                                                          |                  |   |             |
|----------------------------------------------------------|------------------|---|-------------|
| Fixed-effects (within) regression                        | Number of obs    | = | 198         |
| Group variable: <b>entity_id</b>                         | Number of groups | = | 2           |
| R-squared:                                               | Obs per group:   |   |             |
| Within = <b>0.3272</b>                                   | min =            |   | <b>99</b>   |
| Overall = <b>0.2166</b>                                  | avg =            |   | <b>99.0</b> |
|                                                          | max =            |   | <b>99</b>   |
|                                                          | F(1, 1)          | = | .           |
| corr(u_i, Xb) = <b>-0.0000</b>                           | Prob > F         | = | .           |
| (Std. err. adjusted for 2 clusters in <b>entity_id</b> ) |                  |   |             |

| Delinquency_rate       | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|------------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| _5_year_Treasury_yield |             |                                   |        |       |                      |           |
| L1.                    | -.0031339   | .0002303                          | -13.61 | 0.047 | -.0060596            | -.0002082 |
| L2.                    | .0010992    | .000617                           | 1.78   | 0.326 | -.0067408            | .0089393  |
| L3.                    | -.0022712   | .0008138                          | -2.79  | 0.219 | -.0126116            | .0080691  |
| L4.                    | .0078229    | .0002787                          | 28.07  | 0.023 | .0042815             | .0113643  |
| _cons                  | .0152331    | .0037142                          | 4.10   | 0.152 | -.0319606            | .0624267  |
| sigma_u                | .0117044    |                                   |        |       |                      |           |
| sigma_e                | .0096423    |                                   |        |       |                      |           |
| rho                    | .59570755   | (fraction of variance due to u_i) |        |       |                      |           |

31 . xtreg Delinquency\_rate \_5\_year\_Treasury\_yield L(1/4).\_5\_year\_Treasury\_yield, fe vce(cluster entity\_id)

|                                   |                  |   |             |
|-----------------------------------|------------------|---|-------------|
| Fixed-effects (within) regression | Number of obs    | = | 198         |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2           |
| R-squared:                        | Obs per group:   |   |             |
| Within = <b>0.3274</b>            | min =            |   | <b>99</b>   |
| Overall = <b>0.2167</b>           | avg =            |   | <b>99.0</b> |
|                                   | max =            |   | <b>99</b>   |

```

32 . * test dependency of delinquency rate to 10 year Treasury

33 . xtreg Delinquency_rate _10_year_Treasury_yield, fe vce(cluster entity_id)

Fixed-effects (within) regression               Number of obs   =       206
Group variable: entity_id                     Number of groups =        2

R-squared:                                     Obs per group:
    Within = 0.2894                                min =           103
                                                    avg =       103.0
    Overall = 0.1883                                max =           103

corr(u_i, Xb) = -0.0000                        F(1, 1)         =       10.57
                                                    Prob > F         =       0.1900

                                         (Std. err. adjusted for 2 clusters in entity_id)

+-----+-----+-----+-----+-----+-----+
| Delinquency_rate | Robust | t | P>|t| | [95% conf. interval] |
|-----+-----+-----+-----+-----+-----+
| _10_year_Treasury_yield | .0046027 | .0014156 | 3.25 | 0.190 | -.0133848 | .0225902 |
| _cons | .0096425 | .0048393 | 1.99 | 0.296 | -.0518468 | .0711318 |
+-----+-----+-----+-----+-----+-----+
| sigma_u | .01188763 |
| sigma_e | .009742 |
| rho | .59823223 | (fraction of variance due to u_i)

```

```

34 . xtreg Delinquency_rate L(1/4)._10_year_Treasury_yield, fe vce(cluster entity_id)

Fixed-effects (within) regression              Number of obs   =          198
Group variable: entity_id                   Number of groups =           2

R-squared:                                   Obs per group:
    Within = 0.4504                             min =           99
    Overall = 0.2981                             avg =          99.0
                                                    max =           99

                                                    F(1, 1)         =           .
                                                    Prob > F         =           .

                                                    (Std. err. adjusted for 2 clusters in entity_id)

```

| Delinquency_rate               | Coefficient      | Robust<br>std. err.               | t            | P> t         | [95% conf. interval]      |
|--------------------------------|------------------|-----------------------------------|--------------|--------------|---------------------------|
| <b>_10_year_Treasury_yield</b> |                  |                                   |              |              |                           |
| L1.                            | <b>-.0009028</b> | <b>.000596</b>                    | <b>-1.51</b> | <b>0.371</b> | <b>-.0084758 .0066702</b> |
| L2.                            | <b>.0010527</b>  | <b>.0005576</b>                   | <b>1.89</b>  | <b>0.310</b> | <b>-.0060321 .0081376</b> |
| L3.                            | <b>-.0010266</b> | <b>.000617</b>                    | <b>-1.66</b> | <b>0.345</b> | <b>-.0088667 .0068136</b> |
| L4.                            | <b>.0065141</b>  | <b>.0002135</b>                   | <b>30.51</b> | <b>0.021</b> | <b>.0038014 .0092268</b>  |
| <b>_cons</b>                   | <b>.0059828</b>  | <b>.0051875</b>                   | <b>1.15</b>  | <b>0.455</b> | <b>-.0599307 .0718964</b> |
| sigma_u                        | <b>.0117044</b>  |                                   |              |              |                           |
| sigma_e                        | <b>.00871489</b> |                                   |              |              |                           |
| rho                            | <b>.64333396</b> | (fraction of variance due to u_i) |              |              |                           |

```
35 . xtreg Delinquency_rate _10_year_Treasury_yield L(1/4)._10_year_Treasury_yield, fe vce(cluster entity_id)

Fixed-effects (within) regression              Number of obs   =       198
```



```

Group variable: entity_id                Number of groups =      2

R-squared:                                Obs per group:
  Within = 0.4504                          min =      99
  Overall = 0.2981                        avg =     99.0
                                          max =      99

corr(u_i, Xb) = -0.0000                    F(1, 1) = .
                                          Prob > F = .

```

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate        | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|-------------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| _10_year_Treasury_yield |             |                                   |       |       |                      |          |
| ---                     | .0000863    | .0000494                          | 1.75  | 0.331 | -.0005418            | .0007145 |
| L1.                     | -.0010008   | .0006537                          | -1.53 | 0.368 | -.0093069            | .0073052 |
| L2.                     | .0010782    | .0005444                          | 1.98  | 0.298 | -.0058397            | .0079961 |
| L3.                     | -.0010494   | .0006317                          | -1.66 | 0.345 | -.0090757            | .006977  |
| L4.                     | .0065281    | .0002221                          | 29.40 | 0.022 | .0037066             | .0093496 |
| _cons                   | .0059672    | .0051921                          | 1.15  | 0.456 | -.0600041            | .0719385 |
| sigma_u                 | .0117044    |                                   |       |       |                      |          |
| sigma_e                 | .00873763   |                                   |       |       |                      |          |
| rho                     | .64213737   | (fraction of variance due to u_i) |       |       |                      |          |

36 . \* test each rates to the model

37 . xtreg Delinquency\_rate L(3).Real\_GDP\_growth L(2).CPI\_inflation\_rate L(2).Unemployment\_rate L(4).\_3\_month\_Treasury\_rate, fe vce(cluster entity\_id)

```

Fixed-effects (within) regression          Number of obs =      198
Group variable: entity_id                Number of groups =      2

R-squared:                                Obs per group:
  Within = 0.2698                          min =      99
  Overall = 0.1785                        avg =     99.0
                                          max =      99

corr(u_i, Xb) = -0.0000                    F(1, 1) = .
                                          Prob > F = .

```

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate       | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|------------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Real_GDP_growth        |             |                                   |        |       |                      |           |
| L3.                    | -.000151    | .0000111                          | -13.58 | 0.047 | -.0002922            | -9.67e-06 |
| CPI_inflation_rate     |             |                                   |        |       |                      |           |
| L2.                    | -.0003748   | .0000705                          | -5.32  | 0.118 | -.0012701            | .0005205  |
| Unemployment_rate      |             |                                   |        |       |                      |           |
| L2.                    | .0025411    | .0003406                          | 7.46   | 0.085 | -.0017866            | .0068687  |
| _3_month_Treasury_rate |             |                                   |        |       |                      |           |
| L4.                    | .0030721    | .0010738                          | 2.86   | 0.214 | -.0105719            | .0167161  |
| _cons                  | .0064337    | .0040544                          | 1.59   | 0.358 | -.0450824            | .0579497  |
| sigma_u                | .0117044    |                                   |        |       |                      |           |
| sigma_e                | .01004556   |                                   |        |       |                      |           |
| rho                    | .57582791   | (fraction of variance due to u_i) |        |       |                      |           |

38 . xtreg Delinquency\_rate L(3).Real\_GDP\_growth L(2).CPI\_inflation\_rate L(2).Unemployment\_rate L(4).\_5\_year\_Treasury\_yield, fe vce(cluster entity\_id)

```

Fixed-effects (within) regression          Number of obs =      198
Group variable: entity_id                Number of groups =      2

R-squared:                                Obs per group:
  Within = 0.4988                          min =      99
  Overall = 0.3301                        avg =     99.0
                                          max =      99

corr(u_i, Xb) = -0.0000                    F(1, 1) = .
                                          Prob > F = .

```

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate | Coefficient | Robust<br>std. err. | t | P> t | [95% conf. interval] |  |
|------------------|-------------|---------------------|---|------|----------------------|--|
|------------------|-------------|---------------------|---|------|----------------------|--|

|                               |           |                                   |        |       |           |           |
|-------------------------------|-----------|-----------------------------------|--------|-------|-----------|-----------|
| Real_GDP_growth<br>L3.        | -.0000881 | 5.64e-06                          | -15.63 | 0.041 | -.0001598 | -.0000165 |
| CPI_inflation_rate<br>L2.     | -.0001194 | .0001095                          | -1.09  | 0.473 | -.001511  | .0012723  |
| Unemployment_rate<br>L2.      | .0028094  | .0002175                          | 12.92  | 0.049 | .0000455  | .0055733  |
| _5_year_Treasury_yield<br>L4. | .0055468  | .0013052                          | 4.25   | 0.147 | -.0110375 | .022131   |
| _cons                         | -.0056205 | .0051055                          | -1.10  | 0.469 | -.0704915 | .0592506  |
| sigma_u                       | .0117044  |                                   |        |       |           |           |
| sigma_e                       | .00832271 |                                   |        |       |           |           |
| rho                           | .66417485 | (fraction of variance due to u_i) |        |       |           |           |

39 . xtreg Delinquency\_rate L(3).Real\_GDP\_growth L(2).CPI\_inflation\_rate L(2).Unemployment\_rate L(4).\_10\_year\_Treasury\_yield, fe vce(cluster entity\_id)

Fixed-effects (within) regression                      Number of obs       =        198  
Group variable: **entity\_id**                      Number of groups    =        2

R-squared:                                              Obs per group:                      min =        99  
                    Within = **0.5627**                                              avg =        99.0  
                    Overall = **0.3724**                                              max =        99

corr(u\_i, Xb) = **-0.0000**                                              F(1, 1)                      =        .  
                                                                                                 Prob > F                      =        .

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate               | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|--------------------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Real_GDP_growth<br>L3.         | -.0001391   | 9.17e-06                          | -15.18 | 0.042 | -.0002557            | -.0000226 |
| CPI_inflation_rate<br>L2.      | -.0001307   | .0001005                          | -1.30  | 0.417 | -.0014079            | .0011465  |
| Unemployment_rate<br>L2.       | .0018542    | .0000212                          | 87.49  | 0.007 | .0015849             | .0021235  |
| _10_year_Treasury_yield<br>L4. | .0060974    | .0013445                          | 4.54   | 0.138 | -.0109863            | .0231811  |
| _cons                          | -.0054644   | .0046612                          | -1.17  | 0.450 | -.0646909            | .0537621  |
| sigma_u                        | .0117044    |                                   |        |       |                      |           |
| sigma_e                        | .00777355   |                                   |        |       |                      |           |
| rho                            | .69391303   | (fraction of variance due to u_i) |        |       |                      |           |

40 . xtreg Delinquency\_rate L(3).Real\_GDP\_growth L(2).CPI\_inflation\_rate L(2).Unemployment\_rate L(4).BBB\_corporate\_yield, fe vce(cluster entity\_id)

Fixed-effects (within) regression                      Number of obs       =        198  
Group variable: **entity\_id**                      Number of groups    =        2

R-squared:                                              Obs per group:                      min =        99  
                    Within = **0.6993**                                              avg =        99.0  
                    Overall = **0.4627**                                              max =        99

corr(u\_i, Xb) = **-0.0000**                                              F(1, 1)                      =        .  
                                                                                                 Prob > F                      =        .

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate          | Coefficient | Robust<br>std. err. | t     | P> t  | [95% conf. interval] |          |
|---------------------------|-------------|---------------------|-------|-------|----------------------|----------|
| Real_GDP_growth<br>L3.    | -.0001029   | .0000182            | -5.64 | 0.112 | -.0003345            | .0001287 |
| CPI_inflation_rate<br>L2. | -.0000309   | .0000826            | -0.37 | 0.772 | -.0010802            | .0010184 |
| Unemployment_rate<br>L2.  | .0007728    | .0002492            | 3.10  | 0.199 | -.0023937            | .0039394 |
| BBB_corporate_yield       |             |                     |       |       |                      |          |

43 . \* BBB gives the most significant value and better fits

44 . \* test dependency of delinquency rate to Dow Jones

45 . xtreg Delinquency\_rate Dow\_Jones\_Total\_Stock\_Market\_Ind, vce(cluster entity\_id)

```

Random-effects GLS regression           Number of obs   =       206
Group variable:  entity_id             Number of groups  =        2

R-squared:                             Obs per group:
    Within = 0.0000                     min =       103
    Between = 0.0000                    avg =      103.0
    Overall = 0.2058                     max =       103

                                Wald chi2(1)   =       32.86
corr(u_i, X) = 0 (assumed)            Prob > chi2    =       0.0000

                                (Std. err. adjusted for 2 clusters in entity_id)

```

| Delinquency_rate                 | Coefficient | Robust<br>std. err. | z     | P> z  | [95% conf. interval] |                                   |
|----------------------------------|-------------|---------------------|-------|-------|----------------------|-----------------------------------|
| Dow_Jones_Total_Stock_Market_Ind | -4.51e-07   | 7.86e-08            | -5.73 | 0.000 | -6.05e-07            | -2.97e-07                         |
| _cons                            | .0355979    | .0102096            | 3.49  | 0.000 | .0155875             | .0556082                          |
| sigma_u                          | .01185028   |                     |       |       |                      |                                   |
| sigma_e                          | .00955613   |                     |       |       |                      |                                   |
| rho                              | .60595442   |                     |       |       |                      | (fraction of variance due to u_i) |

46 . xtreg Delinquency\_rate L(1/4).Dow\_Jones\_Total\_Stock\_Market\_Ind, vce(cluster entity\_id)

```

Random-effects GLS regression           Number of obs   =       198
Group variable:  entity_id             Number of groups  =        2

R-squared:                             Obs per group:
    Within = 0.0000                     min =        99
    Between = 0.0000                    avg =       99.0
    Overall = 0.2111                     max =        99

                                Wald chi2(1)   =          .
corr(u_i, X) = 0 (assumed)            Prob > chi2    =          .

                                (Std. err. adjusted for 2 clusters in entity_id)

```

| Delinquency_rate                 | Coefficient | Robust<br>std. err. | z      | P> z  | [95% conf. interval] |                                   |
|----------------------------------|-------------|---------------------|--------|-------|----------------------|-----------------------------------|
| Dow_Jones_Total_Stock_Market_Ind |             |                     |        |       |                      |                                   |
| L1.                              | -4.15e-07   | 1.07e-07            | -3.88  | 0.000 | -6.25e-07            | -2.05e-07                         |
| L2.                              | -2.95e-07   | 2.83e-08            | -10.45 | 0.000 | -3.51e-07            | -2.40e-07                         |
| L3.                              | -7.08e-08   | 2.51e-08            | -2.82  | 0.005 | -1.20e-07            | -2.15e-08                         |
| L4.                              | 3.38e-07    | 2.27e-08            | 14.86  | 0.000 | 2.93e-07             | 3.82e-07                          |
| _cons                            | .0352941    | .0100864            | 3.50   | 0.000 | .0155251             | .0550631                          |
| sigma_u                          | .01166372   |                     |        |       |                      |                                   |
| sigma_e                          | .00970136   |                     |        |       |                      |                                   |
| rho                              | .59100062   |                     |        |       |                      | (fraction of variance due to u_i) |

47 . xtreg Delinquency\_rate Dow\_Jones\_Total\_Stock\_Market\_Ind L(1/4).Dow\_Jones\_Total\_Stock\_Market\_Ind, vce(cluster entity\_id)

```

Random-effects GLS regression           Number of obs   =       198
Group variable:  entity_id             Number of groups  =        2

R-squared:                             Obs per group:
    Within = 0.0000                     min =        99
    Between = 0.0000                    avg =       99.0
    Overall = 0.2112                     max =        99

                                Wald chi2(1)   =          .
corr(u_i, X) = 0 (assumed)            Prob > chi2    =          .

                                (Std. err. adjusted for 2 clusters in entity_id)

```

| Delinquency_rate                 | Coefficient | Robust<br>std. err. | z      | P> z  | [95% conf. interval] |           |
|----------------------------------|-------------|---------------------|--------|-------|----------------------|-----------|
| Dow_Jones_Total_Stock_Market_Ind |             |                     |        |       |                      |           |
| --.                              | -6.55e-08   | 9.14e-08            | -0.72  | 0.473 | -2.45e-07            | 1.14e-07  |
| L1.                              | -3.48e-07   | 1.42e-08            | -24.55 | 0.000 | -3.76e-07            | -3.20e-07 |
| L2.                              | -2.96e-07   | 2.73e-08            | -10.84 | 0.000 | -3.49e-07            | -2.42e-07 |
| L3.                              | -7.23e-08   | 2.30e-08            | -3.15  | 0.002 | -1.17e-07            | -2.73e-08 |
| L4.                              | 3.41e-07    | 1.77e-08            | 19.27  | 0.000 | 3.07e-07             | 3.76e-07  |
| _cons                            | .0352747    | .0100857            | 3.50   | 0.000 | .0155072             | .0550423  |

|         |                                             |
|---------|---------------------------------------------|
| sigma_u | .01166352                                   |
| sigma_e | .00972572                                   |
| rho     | .58985929 (fraction of variance due to u_i) |

48 . \* test dependency of delinquency rate to House Price

49 . xtreg Delinquency\_rate House\_Price\_Index\_\_Level\_, vce(cluster entity\_id)

Random-effects GLS regression                      Number of obs       =        206  
Group variable: **entity\_id**                      Number of groups    =        2

R-squared:                                              Obs per group:                      min =        103  
Within = **0.0000**                                              avg =        103.0  
Between = **0.0000**                                              max =        103  
Overall = **0.1930**

Wald chi2(1)       =        96.53  
corr(u\_i, X) = 0 (assumed)                      Prob > chi2       =        0.0000

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate                   | Coefficient                                 | Robust<br>std. err. | z             | P> z           | [95% conf. interval]  |                      |
|------------------------------------|---------------------------------------------|---------------------|---------------|----------------|-----------------------|----------------------|
| House_Price_Index__Level_<br>_cons | -.0001025<br>.0445713                       | .0000104<br>.01038  | -9.83<br>4.29 | 0.000<br>0.000 | -.0001229<br>.0242269 | -.000082<br>.0649158 |
| sigma_u                            | .01184921                                   |                     |               |                |                       |                      |
| sigma_e                            | .00969191                                   |                     |               |                |                       |                      |
| rho                                | .59915337 (fraction of variance due to u_i) |                     |               |                |                       |                      |

50 . xtreg Delinquency\_rate L(1/4).House\_Price\_Index\_\_Level\_, vce(cluster entity\_id)

Random-effects GLS regression                      Number of obs       =        198  
Group variable: **entity\_id**                      Number of groups    =        2

R-squared:                                              Obs per group:                      min =        99  
Within = **0.0000**                                              avg =        99.0  
Between = **0.0000**                                              max =        99  
Overall = **0.2405**

Wald chi2(1)       =        .  
corr(u\_i, X) = 0 (assumed)                      Prob > chi2       =        .

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate                 | Coefficient                                 | Robust<br>std. err. | z      | P> z  | [95% conf. interval] |           |
|----------------------------------|---------------------------------------------|---------------------|--------|-------|----------------------|-----------|
| House_Price_Index__Level_<br>L1. | -.0003371                                   | .0005453            | -0.62  | 0.536 | -.0014059            | .0007318  |
| L2.                              | .0002705                                    | .0004408            | 0.61   | 0.539 | -.0005933            | .0011344  |
| L3.                              | -.0008575                                   | .0000678            | -12.65 | 0.000 | -.0009903            | -.0007247 |
| L4.                              | .0008494                                    | .0000454            | 18.72  | 0.000 | .0007605             | .0009384  |
| _cons                            | .0415699                                    | .0082989            | 5.01   | 0.000 | .0253044             | .0578355  |
| sigma_u                          | .01166638                                   |                     |        |       |                      |           |
| sigma_e                          | .00937944                                   |                     |        |       |                      |           |
| rho                              | .60739667 (fraction of variance due to u_i) |                     |        |       |                      |           |

51 . xtreg Delinquency\_rate House\_Price\_Index\_\_Level\_ L(1/4).House\_Price\_Index\_\_Level\_, vce(cluster entity\_id)

Random-effects GLS regression                      Number of obs       =        198  
Group variable: **entity\_id**                      Number of groups    =        2

R-squared:                                              Obs per group:                      min =        99  
Within = **0.0000**                                              avg =        99.0  
Between = **0.0000**                                              max =        99  
Overall = **0.2414**

Wald chi2(1)       =        .  
corr(u\_i, X) = 0 (assumed)                      Prob > chi2       =        .

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate                 | Coefficient | Robust<br>std. err. | z     | P> z  | [95% conf. interval] |          |
|----------------------------------|-------------|---------------------|-------|-------|----------------------|----------|
| House_Price_Index__Level_<br>--. | -.0002389   | .0005067            | -0.47 | 0.637 | -.001232             | .0007541 |

|         |           |                                   |       |       |           |           |
|---------|-----------|-----------------------------------|-------|-------|-----------|-----------|
| L1.     | .0001306  | .0004451                          | 0.29  | 0.769 | -.0007418 | .0010031  |
| L2.     | .0000104  | .0001097                          | 0.09  | 0.924 | -.0002046 | .0002254  |
| L3.     | -.000813  | .0001623                          | -5.01 | 0.000 | -.0011311 | -.0004948 |
| L4.     | .0008357  | .0000164                          | 50.85 | 0.000 | .0008035  | .0008679  |
| _cons   | .0417196  | .0086379                          | 4.83  | 0.000 | .0247896  | .0586496  |
| sigma_u | .01166626 |                                   |       |       |           |           |
| sigma_e | .00939435 |                                   |       |       |           |           |
| rho     | .60663377 | (fraction of variance due to u_i) |       |       |           |           |

52 . \* Index is a leading indicator, so forget about testing lagging.

53 . \* test each rates to the model

54 . xtreg Delinquency\_rate L(3).Real\_GDP\_growth L(2).CPI\_inflation\_rate L(2).Unemployment\_rate L(4).BBB\_corporate\_yield Dow\_Jones\_Total\_Stock\_Market\_Ind, fe vce(cluster > entity\_id)

```

Fixed-effects (within) regression      Number of obs   =      198
Group variable: entity_id             Number of groups =       2

R-squared:                            Obs per group:
    Within = 0.7246                    min =      99
                                         avg =     99.0
    Overall = 0.4795                    max =      99

                                         F(1, 1)         =      .
corr(u_i, Xb) = 0.0000                 Prob > F         =      .

                                         (Std. err. adjusted for 2 clusters in entity_id)

```

| Delinquency_rate                          | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|-------------------------------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Real_GDP_growth<br>L3.                    | -.000117    | .0000245                          | -4.77 | 0.131 | -.0004283            | .0001944 |
| CPI_inflation_rate<br>L2.                 | .0000331    | .000111                           | 0.30  | 0.816 | -.0013775            | .0014437 |
| Unemployment_rate<br>L2.                  | .000425     | .0004033                          | 1.05  | 0.483 | -.0046988            | .0055488 |
| BBB_corporate_yield<br>L4.                | .0053358    | .0005949                          | 8.97  | 0.071 | -.0022231            | .0128947 |
| Dow_Jones_Total_Stock_Market_Ind<br>_cons | -1.54e-07   | 6.81e-08                          | -2.27 | 0.264 | -1.02e-06            | 7.10e-07 |
|                                           | -.0014648   | .000535                           | -2.74 | 0.223 | -.0082631            | .0053335 |
| sigma_u                                   | .0117044    |                                   |       |       |                      |          |
| sigma_e                                   | .00618506   |                                   |       |       |                      |          |
| rho                                       | .78170958   | (fraction of variance due to u_i) |       |       |                      |          |

55 . xtreg Delinquency\_rate L(3).Real\_GDP\_growth L(2).CPI\_inflation\_rate L(2).Unemployment\_rate L(4).BBB\_corporate\_yield House\_Price\_Index\_Level\_, fe vce(cluster entity > \_id)

```

Fixed-effects (within) regression      Number of obs   =      198
Group variable: entity_id             Number of groups =       2

R-squared:                            Obs per group:
    Within = 0.7140                    min =      99
                                         avg =     99.0
    Overall = 0.4725                    max =      99

                                         F(1, 1)         =      .
corr(u_i, Xb) = 0.0000                 Prob > F         =      .

                                         (Std. err. adjusted for 2 clusters in entity_id)

```

| Delinquency_rate           | Coefficient | Robust<br>std. err. | t     | P> t  | [95% conf. interval] |          |
|----------------------------|-------------|---------------------|-------|-------|----------------------|----------|
| Real_GDP_growth<br>L3.     | -.0001193   | .0000206            | -5.79 | 0.109 | -.0003809            | .0001423 |
| CPI_inflation_rate<br>L2.  | .0000849    | .0000991            | 0.86  | 0.549 | -.0011743            | .0013441 |
| Unemployment_rate<br>L2.   | .0004014    | .0003022            | 1.33  | 0.411 | -.003438             | .0042407 |
| BBB_corporate_yield<br>L4. | .0054623    | .0007955            | 6.87  | 0.092 | -.0046449            | .0155695 |

|                          |           |                                   |       |       |           |          |
|--------------------------|-----------|-----------------------------------|-------|-------|-----------|----------|
| House_Price_Index_Level_ | -.0000308 | 4.34e-06                          | -7.10 | 0.089 | -.000086  | .0000243 |
| _cons                    | .0001995  | .0018083                          | 0.11  | 0.930 | -.0227766 | .0231757 |
| <hr/>                    |           |                                   |       |       |           |          |
| sigma_u                  | .0117044  |                                   |       |       |           |          |
| sigma_e                  | .00630287 |                                   |       |       |           |          |
| rho                      | .77520195 | (fraction of variance due to u_i) |       |       |           |          |

```
56 . xtreg Delinquency_rate L(3).Real_GDP_growth L(2).CPI_inflation_rate L(2).Unemployment_rate L(4).BBB_corporate_yield Commercial_Real_Estate_Price_Ind, fe vce(cluster
> entity_id)
```

```
Fixed-effects (within) regression      Number of obs   =    198
Group variable: entity_id             Number of groups =     2
```

|                         |                   |
|-------------------------|-------------------|
| R-squared:              | Obs per group:    |
| Within = <b>0.7124</b>  | min = <b>99</b>   |
|                         | avg = <b>99.0</b> |
| Overall = <b>0.4714</b> | max = <b>99</b>   |

|                        |                                   |   |   |
|------------------------|-----------------------------------|---|---|
| corr(u_i, Xb) = 0.0000 | $\frac{F(1, 1)}{\text{Prob} > F}$ | = | . |
|                        |                                   | = | . |

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate                          | Coefficient           | Robust<br>std. err.               | t             | P> t           | [95% conf. interval]   |                      |
|-------------------------------------------|-----------------------|-----------------------------------|---------------|----------------|------------------------|----------------------|
| Real_GDP_growth<br>L3.                    | -.0001336             | .0000148                          | -9.03         | 0.070          | -.0003214              | .0000543             |
| CPI_inflation_rate<br>L2.                 | .0000536              | .0000732                          | 0.73          | 0.597          | -.0008762              | .0009835             |
| Unemployment_rate<br>L2.                  | .0003305              | .0001996                          | 1.66          | 0.346          | -.0022051              | .0028661             |
| BBB_corporate_yield<br>L4.                | .0050391              | .0009663                          | 5.21          | 0.121          | -.0072389              | .0173172             |
| Commercial_Real_Estate_Price_Ind<br>_cons | -.0000371<br>.0056555 | 4.21e-06<br>.0050376              | -8.79<br>1.12 | 0.072<br>0.463 | -.0000906<br>-.0583528 | .0000165<br>.0696639 |
| sigma_u                                   | .0117044              |                                   |               |                |                        |                      |
| sigma_e                                   | .00632072             |                                   |               |                |                        |                      |
| rho                                       | .77421458             | (fraction of variance due to u_i) |               |                |                        |                      |

```
57 . xtreg Delinquency_rate L(3).Real_GDP_growth L(2).CPI_inflation_rate L(2).Unemployment_rate L(4).BBB_corporate_yield Market_Volatility_Index_Level_, fe vce(cluster
> entity_id)
```

|                                   |                  |   |     |
|-----------------------------------|------------------|---|-----|
| Fixed-effects (within) regression | Number of obs    | = | 198 |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2   |

|                         |                   |
|-------------------------|-------------------|
| R-squared:              | Obs per group:    |
| Within = <b>0.7155</b>  | min = <b>99</b>   |
|                         | avg = <b>99.0</b> |
| Overall = <b>0.4735</b> | max = <b>99</b>   |

|                                |                |   |   |
|--------------------------------|----------------|---|---|
| corr(u_i, Xb) = <b>-0.0000</b> | <u>F(1, 1)</u> | = | . |
|                                | Prob > F       | = | . |

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate                        | Coefficient           | Robust<br>std. err.               | t             | P> t           | [95% conf. interval]   |                      |
|-----------------------------------------|-----------------------|-----------------------------------|---------------|----------------|------------------------|----------------------|
| Real_GDP_growth<br>L3.                  | -.0000705             | .0000226                          | -3.12         | 0.197          | -.0003575              | .0002164             |
| CPI_inflation_rate<br>L2.               | -.0001119             | .0000936                          | -1.20         | 0.443          | -.0013011              | .0010773             |
| Unemployment_rate<br>L2.                | .0007403              | .0002455                          | 3.02          | 0.204          | -.0023793              | .0038599             |
| BBB_corporate_yield<br>L4.              | .0057887              | .0008841                          | 6.55          | 0.096          | -.0054451              | .0170226             |
| Market_Volatility_Index_Level_<br>_cons | .0001203<br>-.0123253 | .000016<br>.0029482               | 7.50<br>-4.18 | 0.084<br>0.149 | -.0000836<br>-.0497852 | .0003242<br>.0251346 |
| sigma_u                                 | .0117044              |                                   |               |                |                        |                      |
| sigma_e                                 | .00628657             |                                   |               |                |                        |                      |
| rho                                     | .77610287             | (fraction of variance due to u_i) |               |                |                        |                      |

```

58 . * market volatility index gives highest significant value and better fits
59 . * inflation rate should be coincidence/lagging. test L1 if it is better using lagging.
60 . xtreg Delinquency_rate L(3).Real_GDP_growth L(1).CPI_inflation_rate L(2).Unemployment_rate L(4).BBB_corporate_yield Market_Volatility_Index__Level_, fe vce(cluster
> entity_id)

```

```

Fixed-effects (within) regression      Number of obs   =      198
Group variable: entity_id            Number of groups =       2

R-squared:                             Obs per group:
  Within = 0.7163                      min =      99
  Overall = 0.4740                     avg =     99.0
                                         max =      99

corr(u_i, Xb) = 0.0000                  F(1, 1)         =      .
                                         Prob > F         =      .

                                         (Std. err. adjusted for 2 clusters in entity_id)

```

| Delinquency_rate                        | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|-----------------------------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Real_GDP_growth<br>L3.                  | -.0000678   | .0000198                          | -3.42  | 0.181 | -.0003192            | .0001837  |
| CPI_inflation_rate<br>L1.               | -.0001682   | 9.45e-06                          | -17.79 | 0.036 | -.0002883            | -.0000481 |
| Unemployment_rate<br>L2.                | .0007664    | .0002787                          | 2.75   | 0.222 | -.002775             | .0043079  |
| BBB_corporate_yield<br>L4.              | .0057653    | .0008588                          | 6.71   | 0.094 | -.0051469            | .0166775  |
| Market_Volatility_Index_Level_<br>_cons | .0001202    | .0000136                          | 8.84   | 0.072 | -.0000526            | .000293   |
|                                         | -.0122137   | .0024369                          | -5.01  | 0.125 | -.0431781            | .0187507  |
| sigma_u                                 | .0117044    |                                   |        |       |                      |           |
| sigma_e                                 | .00627844   |                                   |        |       |                      |           |
| rho                                     | .7765254    | (fraction of variance due to u_i) |        |       |                      |           |

```

61 . * test dependency of delinquency rate to Euro GDP

```

```

62 . xtreg Delinquency_rate Euro_area_real_GDP_growth, fe vce(cluster entity_id)

```

```

Fixed-effects (within) regression      Number of obs   =      206
Group variable: entity_id            Number of groups =       2

R-squared:                             Obs per group:
  Within = 0.0079                      min =     103
  Overall = 0.0051                     avg =    103.0
                                         max =     103

corr(u_i, Xb) = 0.0000                  F(1, 1)         =      3.14
                                         Prob > F         =     0.3270

                                         (Std. err. adjusted for 2 clusters in entity_id)

```

| Delinquency_rate                   | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |          |
|------------------------------------|-------------|-----------------------------------|--------|-------|----------------------|----------|
| Euro_area_real_GDP_growth<br>_cons | -.0001447   | .0000816                          | -1.77  | 0.327 | -.0011817            | .0008923 |
|                                    | .025596     | .0001237                          | 206.94 | 0.003 | .0240243             | .0271676 |
| sigma_u                            | .01188763   |                                   |        |       |                      |          |
| sigma_e                            | .01151071   |                                   |        |       |                      |          |
| rho                                | .51610482   | (fraction of variance due to u_i) |        |       |                      |          |

```

63 . xtreg Delinquency_rate L(1/4).Euro_area_real_GDP_growth, fe vce(cluster entity_id)

```

```

Fixed-effects (within) regression      Number of obs   =      198
Group variable: entity_id            Number of groups =       2

R-squared:                             Obs per group:
  Within = 0.0923                      min =      99
  Overall = 0.0611                     avg =     99.0
                                         max =      99

corr(u_i, Xb) = -0.0000                  F(1, 1)         =      .
                                         Prob > F         =      .

```



(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate          | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|---------------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Euro_area_real_GDP_growth |             |                                   |        |       |                      |           |
| L1.                       | -.0002649   | .0000577                          | -4.59  | 0.137 | -.000998             | .0004683  |
| L2.                       | -.0003701   | .0000552                          | -6.71  | 0.094 | -.0010715            | .0003312  |
| L3.                       | -.0003754   | .0000392                          | -9.57  | 0.066 | -.0008737            | .0001229  |
| L4.                       | -.0002581   | 7.92e-06                          | -32.61 | 0.020 | -.0003587            | -.0001575 |
| _cons                     | .0269475    | .0002349                          | 114.70 | 0.006 | .0239623             | .0299326  |
| sigma_u                   | .0117044    |                                   |        |       |                      |           |
| sigma_e                   | .01119984   |                                   |        |       |                      |           |
| rho                       | .52201845   | (fraction of variance due to u_i) |        |       |                      |           |

```
64 . xtreg Delinquency_rate Euro_area_real_GDP_growth L(1/4).Euro_area_real_GDP_growth, fe vce(cluster entity_id)
```

|                                   |                  |   |             |
|-----------------------------------|------------------|---|-------------|
| Fixed-effects (within) regression | Number of obs    | = | 198         |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2           |
| R-squared:                        | Obs per group:   |   |             |
| Within = <b>0.1126</b>            | min =            |   | <b>99</b>   |
|                                   | avg =            |   | <b>99.0</b> |
| Overall = <b>0.0745</b>           | max =            |   | <b>99</b>   |
|                                   | <u>F(1, 1)</u>   | = | .           |
| corr(u_i, Xb) = <b>-0.0000</b>    | Prob > F         | = |             |

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate          | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|---------------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Euro_area_real_GDP_growth |             |                                   |        |       |                      |           |
| ---                       | -.0002385   | .0001114                          | -2.14  | 0.278 | -.001654             | .0011769  |
| L1.                       | -.0003277   | .0000872                          | -3.76  | 0.166 | -.0014359            | .0007804  |
| L2.                       | -.0004014   | .0000699                          | -5.74  | 0.110 | -.0012899            | .0004871  |
| L3.                       | -.00037     | .0000368                          | -10.05 | 0.063 | -.0008379            | .0000978  |
| L4.                       | -.0002569   | 7.37e-06                          | -34.84 | 0.018 | -.0003506            | -.0001632 |
| _cons                     | .0274182    | .0004554                          | 60.21  | 0.011 | .0216317             | .0332047  |
| sigma_u                   | .0117044    |                                   |        |       |                      |           |
| sigma_e                   | .01110304   |                                   |        |       |                      |           |
| rho                       | .52634854   | (fraction of variance due to u_i) |        |       |                      |           |

65 . \* test dependency of delinquency rate to Developing Asia GDP

```
66 . xtreg Delinquency_rate Developing_Asia_real_GDP_growth, fe vce(cluster entity_id)
```

|                                   |                  |   |               |
|-----------------------------------|------------------|---|---------------|
| Fixed-effects (within) regression | Number of obs    | = | 206           |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2             |
| R-squared:                        | Obs per group:   |   |               |
| Within = <b>0.0217</b>            | min =            |   | <b>103</b>    |
|                                   | avg =            |   | <b>103.0</b>  |
| Overall = <b>0.0141</b>           | max =            |   | <b>103</b>    |
|                                   | F(1, 1)          | = | <b>72.83</b>  |
| corr(u_i, Xb) = <b>0.0000</b>     | Prob > F         | = | <b>0.0743</b> |

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate                         | Coefficient         | Robust<br>std. err.               | t             | P> t           | [95% conf. interval]  |                     |
|------------------------------------------|---------------------|-----------------------------------|---------------|----------------|-----------------------|---------------------|
| Developing_Asia_real_GDP_growth<br>_cons | .000323<br>.0230864 | .0000379<br>.0002684              | 8.53<br>86.02 | 0.074<br>0.007 | -.0001579<br>.0196764 | .000804<br>.0264963 |
| sigma_u                                  | .01188763           |                                   |               |                |                       |                     |
| sigma_e                                  | .01143063           |                                   |               |                |                       |                     |
| rho                                      | .5195911            | (fraction of variance due to u_i) |               |                |                       |                     |

```
67 . xtreg Delinquency_rate L(1/4).Developing_Asia_real_GDP_growth, fe vce(cluster entity_id)
```

```
Fixed-effects (within) regression      Number of obs   =      198
Group variable:  entity_id          Number of groups =        2

R-squared:                                Obs per group:

```

Within = **0.0199** min = **99**  
 Overall = **0.0132** avg = **99.0**  
 max = **99**

corr(u\_i, Xb) = **-0.0000** F(1, 1) = **.**  
 Prob > F = **.**

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate                | Coefficient | Robust<br>std. err.               | t    | P> t  | [95% conf. interval] |          |
|---------------------------------|-------------|-----------------------------------|------|-------|----------------------|----------|
| Developing_Asia_real_GDP_growth |             |                                   |      |       |                      |          |
| L1.                             | .0002368    | .0000557                          | 4.25 | 0.147 | -.0004707            | .0009443 |
| L2.                             | .0001757    | .0000893                          | 1.97 | 0.300 | -.0009595            | .0013109 |
| L3.                             | .0000813    | .0000862                          | 0.94 | 0.519 | -.0010139            | .0011764 |
| L4.                             | .0000819    | .0001093                          | 0.75 | 0.591 | -.0013065            | .0014703 |
| _cons                           | .0209643    | .0024331                          | 8.62 | 0.074 | -.0099508            | .0518795 |
| sigma_u                         | .0117044    |                                   |      |       |                      |          |
| sigma_e                         | .01163823   |                                   |      |       |                      |          |
| rho                             | .50283473   | (fraction of variance due to u_i) |      |       |                      |          |

68 . xtreg Delinquency\_rate Developing\_Asia\_real\_GDP\_growth L(1/4).Developing\_Asia\_real\_GDP\_growth, fe vce(cluster entity\_id)

Fixed-effects (within) regression Number of obs = **198**  
 Group variable: **entity\_id** Number of groups = **2**

R-squared: Obs per group:  
 Within = **0.0458** min = **99**  
 Overall = **0.0303** avg = **99.0**  
 max = **99**

corr(u\_i, Xb) = **0.0000** F(1, 1) = **.**  
 Prob > F = **.**

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate                | Coefficient | Robust<br>std. err.               | t    | P> t  | [95% conf. interval] |          |
|---------------------------------|-------------|-----------------------------------|------|-------|----------------------|----------|
| Developing_Asia_real_GDP_growth |             |                                   |      |       |                      |          |
| --.                             | .0003528    | .0000559                          | 6.31 | 0.100 | -.000358             | .0010635 |
| L1.                             | .0002408    | .0000565                          | 4.26 | 0.147 | -.0004765            | .0009581 |
| L2.                             | .0001765    | .0000897                          | 1.97 | 0.299 | -.0009633            | .0013162 |
| L3.                             | .0001234    | .0000931                          | 1.33 | 0.411 | -.0010595            | .0013064 |
| L4.                             | .0000978    | .0001121                          | 0.87 | 0.543 | -.0013263            | .0015219 |
| _cons                           | .0180057    | .0029085                          | 6.19 | 0.102 | -.0189509            | .0549623 |
| sigma_u                         | .0117044    |                                   |      |       |                      |          |
| sigma_e                         | .0115137    |                                   |      |       |                      |          |
| rho                             | .50821298   | (fraction of variance due to u_i) |      |       |                      |          |

69 . \* test dependency of delinquency rate to Japan GDP

70 . xtreg Delinquency\_rate Japan\_real\_GDP\_growth, fe vce(cluster entity\_id)

Fixed-effects (within) regression Number of obs = **206**  
 Group variable: **entity\_id** Number of groups = **2**

R-squared: Obs per group:  
 Within = **0.0030** min = **103**  
 Overall = **0.0020** avg = **103.0**  
 max = **103**

corr(u\_i, Xb) = **-0.0000** F(1, 1) = **1.05**  
 Prob > F = **0.4928**

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate      | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |          |
|-----------------------|-------------|-----------------------------------|--------|-------|----------------------|----------|
| Japan_real_GDP_growth | -.0001251   | .0001224                          | -1.02  | 0.493 | -.0016798            | .0014296 |
| _cons                 | .0254856    | .0001064                          | 239.44 | 0.003 | .0241331             | .026838  |
| sigma_u               | .01188763   |                                   |        |       |                      |          |
| sigma_e               | .01153898   |                                   |        |       |                      |          |
| rho                   | .51487935   | (fraction of variance due to u_i) |        |       |                      |          |

```
74 . xtreg Delinquency_rate U_K_real_GDP_growth, fe vce(cluster entity_id)
```

```
Fixed-effects (within) regression      Number of obs   =      206
Group variable: entity_id             Number of groups =       2
```

|                  |                |
|------------------|----------------|
| R-squared:       | Obs per group: |
| Within = 0.0072  | min = 103      |
|                  | avg = 103.0    |
| Overall = 0.0047 | max = 103      |

|                        |          |   |        |
|------------------------|----------|---|--------|
|                        | F(1, 1)  | = | 2.02   |
| corr(u_i, Xb) = 0.0000 | Prob > F | = | 0.3902 |

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate      | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|-----------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Japan_real_GDP_growth |             |                                   |        |       |                      |           |
| —.                    | -.000307    | .0001535                          | -2.00  | 0.295 | -.0022576            | .0016436  |
| L1.                   | -.0003302   | .0001232                          | -2.68  | 0.227 | -.0018961            | .0012358  |
| L2.                   | -.0003391   | .0000586                          | -5.79  | 0.109 | -.0010832            | .0004049  |
| L3.                   | -.0005006   | .0000334                          | -15.00 | 0.042 | -.0009247            | -.0000765 |
| L4.                   | -.0004382   | .0000113                          | -38.82 | 0.016 | -.0005816            | -.0002948 |
| _cons                 | .026642     | .000303                           | 87.92  | 0.007 | .0227915             | .0304925  |
| sigma_u               | .0117044    |                                   |        |       |                      |           |
| sigma_e               | .01106964   |                                   |        |       |                      |           |
| rho                   | .52785042   | (fraction of variance due to u_i) |        |       |                      |           |

```
74 . xtreg Delinquency_rate U_K__real_GDP_growth, fe vce(cluster entity_id)
```

|                  |                |
|------------------|----------------|
| R-squared:       | Obs per group: |
| Within = 0.0072  | min = 103      |
|                  | avg = 103.0    |
| Overall = 0.0047 | max = 103      |

(Std. err. adjusted for 2 clusters in **entity\_id**)

|                     |          |          |        |       |           |          |
|---------------------|----------|----------|--------|-------|-----------|----------|
| U_K_real_GDP_growth | -.000088 | .0000619 | -1.42  | 0.390 | -.0008744 | .0006984 |
| _cons               | .0255669 | .0001338 | 191.14 | 0.003 | .0238673  | .0272665 |

|         |           |                                   |
|---------|-----------|-----------------------------------|
| sigma_u | .01188763 |                                   |
| sigma_e | .01151508 |                                   |
| rho     | .51591522 | (fraction of variance due to u_i) |

75 . xtreg Delinquency\_rate L(1/4).U\_K\_real\_GDP\_growth, fe vce(cluster entity\_id)

Fixed-effects (within) regression                      Number of obs    =        198  
Group variable: **entity\_id**                      Number of groups   =        2

R-squared:                                              Obs per group:                      min =        99  
Within   = **0.1025**                                              avg =        99.0  
Overall   = **0.0678**                                              max =        99

corr(u\_i, Xb) = **-0.0000**                                               $\frac{F(1, 1)}{\text{Prob} > F}$                       =        .

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate    | Coefficient | Robust<br>std. err. | t     | P> t  | [95% conf. interval] |                                   |
|---------------------|-------------|---------------------|-------|-------|----------------------|-----------------------------------|
| U_K_real_GDP_growth |             |                     |       |       |                      |                                   |
| L1.                 | -.0001681   | .0000658            | -2.56 | 0.237 | -.001004             | .0006678                          |
| L2.                 | -.0002772   | .0000883            | -3.14 | 0.196 | -.0013986            | .0008442                          |
| L3.                 | -.0002915   | .0000772            | -3.77 | 0.165 | -.0012729            | .0006899                          |
| L4.                 | -.0001943   | .000037             | -5.25 | 0.120 | -.0006643            | .0002756                          |
| _cons               | .0270804    | .0005779            | 46.86 | 0.014 | .0197369             | .0344239                          |
| sigma_u             | .0117044    |                     |       |       |                      |                                   |
| sigma_e             | .01113713   |                     |       |       |                      |                                   |
| rho                 | .52481962   |                     |       |       |                      | (fraction of variance due to u_i) |

76 . xtreg Delinquency\_rate U\_K\_real\_GDP\_growth L(1/4).U\_K\_real\_GDP\_growth, fe vce(cluster entity\_id)

Fixed-effects (within) regression                      Number of obs    =        198  
Group variable: **entity\_id**                      Number of groups   =        2

R-squared:                                              Obs per group:                      min =        99  
Within   = **0.1200**                                              avg =        99.0  
Overall   = **0.0794**                                              max =        99

corr(u\_i, Xb) = **-0.0000**                                               $\frac{F(1, 1)}{\text{Prob} > F}$                       =        .

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate    | Coefficient | Robust<br>std. err. | t     | P> t  | [95% conf. interval] |                                   |
|---------------------|-------------|---------------------|-------|-------|----------------------|-----------------------------------|
| U_K_real_GDP_growth |             |                     |       |       |                      |                                   |
| _.                  | -.0001478   | .0000941            | -1.57 | 0.361 | -.001343             | .0010473                          |
| L1.                 | -.0002186   | .0000981            | -2.23 | 0.269 | -.0014651            | .0010278                          |
| L2.                 | -.0003019   | .0001042            | -2.90 | 0.212 | -.0016261            | .0010222                          |
| L3.                 | -.0002718   | .0000649            | -4.19 | 0.149 | -.0010961            | .0005525                          |
| L4.                 | -.0001943   | .0000371            | -5.24 | 0.120 | -.0006652            | .0002765                          |
| _cons               | .0275115    | .0008538            | 32.22 | 0.020 | .0166634             | .0383596                          |
| sigma_u             | .0117044    |                     |       |       |                      |                                   |
| sigma_e             | .01105694   |                     |       |       |                      |                                   |
| rho                 | .52842294   |                     |       |       |                      | (fraction of variance due to u_i) |

77 . \* test Euro GDP and Japan GDP to the model

78 . xtreg Delinquency\_rate L(3).Real\_GDP\_growth L(1).CPI\_inflation\_rate L(2).Unemployment\_rate L(4).BBB\_corporate\_yield Market\_Volatility\_Index\_Level\_ L(3).Euro\_area\_r  
> eal\_GDP\_growth, fe vce(cluster entity\_id)

Fixed-effects (within) regression                      Number of obs    =        198  
Group variable: **entity\_id**                      Number of groups   =        2

R-squared:                                              Obs per group:                      min =        99  
Within   = **0.7198**                                              avg =        99.0  
Overall   = **0.4763**                                              max =        99

corr(u\_i, Xb) = **0.0000**                                               $\frac{F(1, 1)}{\text{Prob} > F}$                       =        .

| (Std. err. adjusted for 2 clusters in <b>entity_id</b> ) |             |                                   |        |       |                      |           |
|----------------------------------------------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Deliquency_rate                                          | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
| Real_GDP_growth<br>L3.                                   | -.0003618   | .0000153                          | -23.72 | 0.027 | -.0005556            | -.0001679 |
| CPI_inflation_rate<br>L1.                                | -.0001968   | 6.06e-06                          | -32.44 | 0.020 | -.0002738            | -.0001197 |
| Unemployment_rate<br>L2.                                 | .0007959    | .000283                           | 2.81   | 0.217 | -.0027995            | .0043912  |
| BBB_corporate_yield<br>L4.                               | .0057395    | .0008641                          | 6.64   | 0.095 | -.0052403            | .0167194  |
| Market_Volatility_Index_Level_<br>L3.                    | .0001135    | .0000128                          | 8.84   | 0.072 | -.0000496            | .0002767  |
| Euro_area_real_GDP_growth<br>L3.                         | .0002302    | .0000275                          | 8.38   | 0.076 | -.0001189            | .0005793  |
| _cons                                                    | -.0116528   | .0025103                          | -4.64  | 0.135 | -.0435489            | .0202433  |
| sigma_u                                                  | .0117044    |                                   |        |       |                      |           |
| sigma_e                                                  | .00625544   |                                   |        |       |                      |           |
| rho                                                      | .77782332   | (fraction of variance due to u_i) |        |       |                      |           |

79 . xtreg Deliquency\_rate L(3).Real\_GDP\_growth L(1).CPI\_inflation\_rate L(2).Unemployment\_rate L(4).BBB\_corporate\_yield Market\_Volatility\_Index\_Level\_ L(3).Japan\_real\_ > GDP\_growth, fe vce(cluster entity\_id)

|                                                          |                  |   |      |
|----------------------------------------------------------|------------------|---|------|
| Fixed-effects (within) regression                        | Number of obs    | = | 198  |
| Group variable: <b>entity_id</b>                         | Number of groups | = | 2    |
| R-squared:                                               | Obs per group:   |   |      |
| Within = <b>0.7168</b>                                   | min =            |   | 99   |
| Overall = <b>0.4743</b>                                  | avg =            |   | 99.0 |
|                                                          | max =            |   | 99   |
| corr(u_i, Xb) = <b>0.0000</b>                            | F(1, 1)          | = | .    |
|                                                          | Prob > F         | = | .    |
| (Std. err. adjusted for 2 clusters in <b>entity_id</b> ) |                  |   |      |

| Deliquency_rate                       | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|---------------------------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Real_GDP_growth<br>L3.                | -.0000122   | .0000681                          | -0.18 | 0.887 | -.0008772            | .0008528 |
| CPI_inflation_rate<br>L1.             | -.000147    | .0000279                          | -5.27 | 0.119 | -.0005013            | .0002073 |
| Unemployment_rate<br>L2.              | .0007856    | .0002961                          | 2.65  | 0.229 | -.0029766            | .0045478 |
| BBB_corporate_yield<br>L4.            | .0057604    | .0008653                          | 6.66  | 0.095 | -.0052345            | .0167553 |
| Market_Volatility_Index_Level_<br>L3. | .0001196    | .0000131                          | 9.13  | 0.069 | -.0000468            | .000286  |
| Japan_real_GDP_growth<br>L3.          | -.0000774   | .0000672                          | -1.15 | 0.455 | -.0009317            | .0007769 |
| _cons                                 | -.0123983   | .002283                           | -5.43 | 0.116 | -.0414068            | .0166102 |
| sigma_u                               | .0117044    |                                   |       |       |                      |          |
| sigma_e                               | .006289     |                                   |       |       |                      |          |
| rho                                   | .77596868   | (fraction of variance due to u_i) |       |       |                      |          |

|                                                  |             |                                   |                                  |
|--------------------------------------------------|-------------|-----------------------------------|----------------------------------|
| Within = 0.7213                                  |             | min = 99                          |                                  |
| Overall = 0.4773                                 |             | avg = 99.0                        |                                  |
|                                                  |             | max = 99                          |                                  |
| corr(u_i, Xb) = 0.0000                           |             | F(1, 1)                           | = .                              |
|                                                  |             | Prob > F                          | = .                              |
| (Std. err. adjusted for 2 clusters in entity_id) |             |                                   |                                  |
| Delinquency_rate                                 | Coefficient | Robust std. err.                  | t P> t  [95% conf. interval]     |
| Real_GDP_growth L3.                              | -.0003137   | .0000127                          | -24.68 0.026 -.0004751 -.0001522 |
| CPI_inflation_rate L1.                           | -.0001648   | .0000247                          | -6.68 0.095 -.0004783 .0001486   |
| Unemployment_rate L2.                            | .0008337    | .0003057                          | 2.73 0.224 -.003051 .0047185     |
| BBB_corporate_yield L4.                          | .0057269    | .0008738                          | 6.55 0.096 -.0053754 .0168291    |
| Market_Volatility_Index_Level_                   | .0001114    | .0000116                          | 9.59 0.066 -.0000362 .000259     |
| Euro_area_real_GDP_growth L3.                    | .0002676    | .0000493                          | 5.43 0.116 -.0003589 .0008941    |
| Japan_real_GDP_growth L3.                        | -.0001336   | .0000778                          | -1.72 0.336 -.0011217 .0008545   |
| _cons                                            | -.0118803   | .0023845                          | -4.98 0.126 -.0421779 .0184173   |
| sigma_u                                          | .0117044    |                                   |                                  |
| sigma_e                                          | .00625516   |                                   |                                  |
| rho                                              | .7778389    | (fraction of variance due to u_i) |                                  |

|                                                                            |             |                                   |                                |
|----------------------------------------------------------------------------|-------------|-----------------------------------|--------------------------------|
| 34 . * test dependency of delinquency rate to Euro Inflation               |             |                                   |                                |
| 35 . xtreg Delinquency_rate Euro_area_inflation, fe vce(cluster entity_id) |             |                                   |                                |
| Fixed-effects (within) regression                                          |             | Number of obs                     | = 206                          |
| Group variable: entity_id                                                  |             | Number of groups                  | = 2                            |
| R-squared:                                                                 |             | Obs per group:                    |                                |
| Within = 0.0152                                                            |             | min                               | = 103                          |
|                                                                            |             | avg                               | = 103.0                        |
| Overall = 0.0099                                                           |             | max                               | = 103                          |
|                                                                            |             | F(1, 1)                           | = 7.63                         |
| corr(u_i, Xb) = 0.0000                                                     |             | Prob > F                          | = 0.2211                       |
| (Std. err. adjusted for 2 clusters in entity_id)                           |             |                                   |                                |
| Delinquency_rate                                                           | Coefficient | Robust std. err.                  | t P> t  [95% conf. interval]   |
| Euro_area_inflation                                                        | -.0006682   | .0002419                          | -2.76 0.221 -.0037417 .0024052 |
| _cons                                                                      | .0268164    | .0005211                          | 51.46 0.012 .0201951 .0334376  |
| sigma_u                                                                    | .01188763   |                                   |                                |
| sigma_e                                                                    | .01146856   |                                   |                                |
| rho                                                                        | .51793695   | (fraction of variance due to u_i) |                                |

|                                                                                   |             |                  |                              |
|-----------------------------------------------------------------------------------|-------------|------------------|------------------------------|
| 36 . xtreg Delinquency_rate L(1/4).Euro_area_inflation, fe vce(cluster entity_id) |             |                  |                              |
| Fixed-effects (within) regression                                                 |             | Number of obs    | = 198                        |
| Group variable: entity_id                                                         |             | Number of groups | = 2                          |
| R-squared:                                                                        |             | Obs per group:   |                              |
| Within = 0.0204                                                                   |             | min              | = 99                         |
|                                                                                   |             | avg              | = 99.0                       |
| Overall = 0.0135                                                                  |             | max              | = 99                         |
|                                                                                   |             | F(1, 1)          | = .                          |
| corr(u_i, Xb) = -0.0000                                                           |             | Prob > F         | = .                          |
| (Std. err. adjusted for 2 clusters in entity_id)                                  |             |                  |                              |
| Delinquency_rate                                                                  | Coefficient | Robust std. err. | t P> t  [95% conf. interval] |

|                     |           |                                   |        |       |           |          |
|---------------------|-----------|-----------------------------------|--------|-------|-----------|----------|
| Euro_area_inflation |           |                                   |        |       |           |          |
| L1.                 | -.000628  | .0003614                          | -1.74  | 0.332 | -.0052197 | .0039636 |
| L2.                 | -.0002771 | .0000219                          | -12.67 | 0.050 | -.000555  | 8.77e-07 |
| L3.                 | -.0001227 | .0001694                          | -0.72  | 0.601 | -.0022756 | .0020301 |
| L4.                 | .0004731  | .0004072                          | 1.16   | 0.452 | -.0047004 | .0056466 |
| _cons               | .0262634  | .0004181                          | 62.81  | 0.010 | .0209508  | .0315759 |
| sigma_u             | .0117044  |                                   |        |       |           |          |
| sigma_e             | .01163502 |                                   |        |       |           |          |
| rho                 | .50297259 | (fraction of variance due to u_i) |        |       |           |          |

37 . xtreg Delinquency\_rate Euro\_area\_inflation L(1/4).Euro\_area\_inflation, fe vce(cluster entity\_id)

Fixed-effects (within) regression                      Number of obs       =        198  
Group variable: **entity\_id**                      Number of groups    =        2

R-squared:                                              Obs per group:                      min =        99  
Within   = **0.0220**                                              avg =        99.0  
Overall   = **0.0146**                                              max =        99

corr(u\_i, Xb) = **0.0000**                                              F(1, 1)                      =        .  
Prob > F                                              =        .

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate    | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |          |
|---------------------|-------------|-----------------------------------|--------|-------|----------------------|----------|
| Euro_area_inflation |             |                                   |        |       |                      |          |
| ---                 | -.0003176   | .0003401                          | -0.93  | 0.522 | -.0046387            | .0040035 |
| L1.                 | -.0004324   | .0001528                          | -2.83  | 0.216 | -.0023742            | .0015095 |
| L2.                 | -.0002268   | .0000319                          | -7.10  | 0.009 | -.0006324            | .0001789 |
| L3.                 | -.0000686   | .0002279                          | -0.30  | 0.814 | -.0029642            | .0028271 |
| L4.                 | .0003916    | .000321                           | 1.22   | 0.437 | -.0036865            | .0044697 |
| _cons               | .0264742    | .0001935                          | 136.84 | 0.005 | .0240158             | .0289325 |
| sigma_u             | .0117044    |                                   |        |       |                      |          |
| sigma_e             | .01165605   |                                   |        |       |                      |          |
| rho                 | .50206987   | (fraction of variance due to u_i) |        |       |                      |          |

38 . \* test dependency of delinquency rate to Asia Inflation

39 . xtreg Delinquency\_rate Developing\_Asia\_inflation, fe vce(cluster entity\_id)

Fixed-effects (within) regression                      Number of obs       =        206  
Group variable: **entity\_id**                      Number of groups    =        2

R-squared:                                              Obs per group:                      min =        103  
Within   = **0.0039**                                              avg =        103.0  
Overall   = **0.0025**                                              max =        103

corr(u\_i, Xb) = **0.0000**                                              F(1, 1)                      =        0.67  
Prob > F                                              =        0.5631

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate          | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|---------------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Developing_Asia_inflation | .0003686    | .0004499                          | 0.82  | 0.563 | -.0053485            | .0060857 |
| _cons                     | .0244363    | .001148                           | 21.29 | 0.030 | .0098493             | .0390232 |
| sigma_u                   | .01188763   |                                   |       |       |                      |          |
| sigma_e                   | .01153401   |                                   |       |       |                      |          |
| rho                       | .51509447   | (fraction of variance due to u_i) |       |       |                      |          |

40 . xtreg Delinquency\_rate L(1/4).Developing\_Asia\_inflation, fe vce(cluster entity\_id)

Fixed-effects (within) regression                      Number of obs       =        198  
Group variable: **entity\_id**                      Number of groups    =        2

R-squared:                                              Obs per group:                      min =        99  
Within   = **0.0019**                                              avg =        99.0  
Overall   = **0.0013**                                              max =        99

corr(u\_i, Xb) = **-0.0000**                                              F(1, 1)                      =        .  
Prob > F                                              =        .

|                           |             | Robust                            |       |       |                      |          |
|---------------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Delinquency_rate          | Coefficient | std. err.                         | t     | P> t  | [95% conf. interval] |          |
| Developing_Asia_inflation |             |                                   |       |       |                      |          |
| L1.                       | .0002482    | .0003413                          | 0.73  | 0.600 | -.0040885            | .0045849 |
| L2.                       | -.0001261   | .0001933                          | -0.65 | 0.632 | -.002582             | .0023298 |
| L3.                       | -.000181    | .0002068                          | -0.88 | 0.542 | -.0028085            | .0024466 |
| L4.                       | .0001522    | .0004542                          | 0.34  | 0.794 | -.0056193            | .0059238 |
| _cons                     | .0248261    | .0031445                          | 7.90  | 0.000 | -.0151285            | .0647808 |
| sigma_u                   | .0117044    |                                   |       |       |                      |          |
| sigma_e                   | .01174445   |                                   |       |       |                      |          |
| rho                       | .49829206   | (fraction of variance due to u_i) |       |       |                      |          |

|                                   |                  |   |      |
|-----------------------------------|------------------|---|------|
| Fixed-effects (within) regression | Number of obs    | = | 198  |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2    |
| R-squared:                        | Obs per group:   |   |      |
| Within = <b>0.0086</b>            | min =            |   | 99   |
|                                   | avg =            |   | 99.0 |
| Overall = <b>0.0057</b>           | max =            |   | 99   |
|                                   | <u>F(1, 1)</u>   | = | .    |
| corr(u_i, Xb) = <b>0.0000</b>     | Prob > F         | = |      |

| Delinquency_rate          | Robust      |                                   |       |       |                      |          |
|---------------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
|                           | Coefficient | std. err.                         | t     | P> t  | [95% conf. interval] |          |
| Developing_Asia_inflation |             |                                   |       |       |                      |          |
| --                        | .0005734    | .0003493                          | 1.64  | 0.348 | -.0038649            | .0050117 |
| L1.                       | -.0000274   | .0001743                          | -0.16 | 0.901 | -.0022416            | .0021868 |
| L2.                       | -.000185    | .0001579                          | -1.17 | 0.450 | -.0021914            | .0018214 |
| L3.                       | -.000204    | .0001933                          | -1.06 | 0.483 | -.00266              | .0022519 |
| L4.                       | .0002394    | .0005085                          | 0.47  | 0.720 | -.0062218            | .0067005 |
| _cons                     | .0240444    | .0036289                          | 6.63  | 0.095 | -.0220656            | .0701543 |
| sigma_u                   | .0117044    |                                   |       |       |                      |          |
| sigma_e                   | .01173595   |                                   |       |       |                      |          |
| rho                       | .49865419   | (fraction of variance due to u_i) |       |       |                      |          |

|                                   |                  |   |               |
|-----------------------------------|------------------|---|---------------|
| Fixed-effects (within) regression | Number of obs    | = | 206           |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2             |
| R-squared:                        | Obs per group:   |   |               |
| Within = <b>0.1756</b>            | min =            |   | <b>103</b>    |
|                                   | avg =            |   | <b>103.0</b>  |
| Overall = <b>0.1143</b>           | max =            |   | <b>103</b>    |
|                                   | F(1, 1)          | = | <b>730.89</b> |
| corr(u_i, Xb) = <b>0.0000</b>     | Prob > F         | = | <b>0.0235</b> |

|                  | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Delinquency_rate |             |                                   |        |       |                      |           |
| Japan_inflation  | -.0026116   | .0000966                          | -27.03 | 0.024 | -.0038391            | -.0013842 |
| _cons            | .0268118    | .0000531                          | 505.08 | 0.001 | .0261373             | .0274863  |
| sigma_u          | .01188763   |                                   |        |       |                      |           |
| sigma_e          | .01049293   |                                   |        |       |                      |           |
| rho              | .56207661   | (fraction of variance due to u_i) |        |       |                      |           |

|                                   |                  |   |     |
|-----------------------------------|------------------|---|-----|
| Fixed-effects (within) regression | Number of obs    | = | 198 |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2   |
| R-squared:                        | Obs per group:   |   |     |



Within = **0.2193** min = **99**  
 Overall = **0.1451** avg = **99.0**  
 max = **99**

corr(u\_i, Xb) = **-0.0000** F(1, 1) = **.**  
 Prob > F = **.**

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Japan_inflation  |             |                                   |        |       |                      |           |
| L1.              | -.0015607   | .000122                           | -12.79 | 0.050 | -.0031106            | -.0000107 |
| L2.              | -.0011172   | .0001174                          | -9.52  | 0.067 | -.0026083            | .0003739  |
| L3.              | -.000817    | .0002049                          | -3.99  | 0.156 | -.0034207            | .0017868  |
| L4.              | -.0003255   | .0001506                          | -2.16  | 0.276 | -.0022385            | .0015875  |
| _cons            | .0271486    | .0001649                          | 164.67 | 0.004 | .0250538             | .0292434  |
| sigma_u          | .0117044    |                                   |        |       |                      |           |
| sigma_e          | .01038702   |                                   |        |       |                      |           |
| rho              | .55942198   | (fraction of variance due to u_i) |        |       |                      |           |

45 . xtreg Delinquency\_rate Japan\_inflation L(1/4).Japan\_inflation, fe vce(cluster entity\_id)

Fixed-effects (within) regression Number of obs = **198**  
 Group variable: **entity\_id** Number of groups = **2**

R-squared: Obs per group:  
 Within = **0.2607** min = **99**  
 Overall = **0.1725** avg = **99.0**  
 max = **99**

corr(u\_i, Xb) = **-0.0000** F(1, 1) = **.**  
 Prob > F = **.**

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Japan_inflation  |             |                                   |        |       |                      |           |
| ---              | -.0014734   | .0001969                          | -7.48  | 0.085 | -.0039756            | .0010287  |
| L1.              | -.0011385   | .0000659                          | -17.28 | 0.037 | -.0019756            | -.0003014 |
| L2.              | -.0008258   | .0001566                          | -5.27  | 0.119 | -.0028156            | .0011639  |
| L3.              | -.000519    | .0002453                          | -2.12  | 0.281 | -.0036355            | .0025974  |
| L4.              | -.0004175   | .0001387                          | -3.01  | 0.204 | -.0021793            | .0013443  |
| _cons            | .0275231    | .0001152                          | 238.84 | 0.003 | .0260589             | .0289874  |
| sigma_u          | .0117044    |                                   |        |       |                      |           |
| sigma_e          | .0101343    |                                   |        |       |                      |           |
| rho              | .57152595   | (fraction of variance due to u_i) |        |       |                      |           |

46 . \* test dependency of delinquency rate to UK Inflation

47 . xtreg Delinquency\_rate U\_K\_inflation, fe vce(cluster entity\_id)

Fixed-effects (within) regression Number of obs = **206**  
 Group variable: **entity\_id** Number of groups = **2**

R-squared: Obs per group:  
 Within = **0.0272** min = **103**  
 Overall = **0.0177** avg = **103.0**  
 max = **103**

corr(u\_i, Xb) = **0.0000** F(1, 1) = **26.05**  
 Prob > F = **0.1232**

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_r~e | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|-----------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| U_K_inflation   | -.0007796   | .0001527                          | -5.10 | 0.123 | -.0027205            | .0011612 |
| _cons           | .0273992    | .0003963                          | 69.15 | 0.009 | .0223643             | .032434  |
| sigma_u         | .01188763   |                                   |       |       |                      |          |
| sigma_e         | .01139827   |                                   |       |       |                      |          |
| rho             | .52100617   | (fraction of variance due to u_i) |       |       |                      |          |

```
48 . xtreg Delinquency_rate L(1/4).U_K_inflation, fe vce(cluster entity_id)
```

```
Fixed-effects (within) regression      Number of obs   =    198
Group variable: entity_id             Number of groups =     2
```

|                         |                   |
|-------------------------|-------------------|
| R-squared:              | Obs per group:    |
| Within = <b>0.0251</b>  | min = <b>99</b>   |
|                         | avg = <b>99.0</b> |
| Overall = <b>0.0166</b> | max = <b>99</b>   |

|                                     |           |   |   |
|-------------------------------------|-----------|---|---|
|                                     | $F(1, 1)$ | = | . |
| <code>corr(u_i, Xb) = 0.0000</code> | Prob > F  | = | . |

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_r~e | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|-----------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| U_K_inflation   |             |                                   |       |       |                      |          |
| L1.             | -.0004988   | .0002421                          | -2.06 | 0.288 | -.0035745            | .0025769 |
| L2.             | -.0002818   | .0000505                          | -5.58 | 0.113 | -.0009238            | .0003601 |
| L3.             | -.0000165   | .0002541                          | -0.06 | 0.959 | -.0032454            | .0032124 |
| L4.             | -.0000283   | .0002655                          | -0.11 | 0.932 | -.0034024            | .0033458 |
| _cons           | .0272395    | .0005594                          | 48.69 | 0.013 | .020131              | .0343479 |
| sigma_u         | .0117044    |                                   |       |       |                      |          |
| sigma_e         | .01160733   |                                   |       |       |                      |          |
| rho             | .50416409   | (fraction of variance due to u_i) |       |       |                      |          |

```
49 . xtreg Delinquency_rate U_K__inflation L(1/4).U_K__inflation, fe vce(cluster entity_id)
```

```
Fixed-effects (within) regression      Number of obs   =    198
Group variable: entity_id             Number of groups =     2
```

|                  |                |
|------------------|----------------|
| R-squared:       | Obs per group: |
| Within = 0.0282  | min = 99       |
|                  | avg = 99.0     |
| Overall = 0.0186 | max = 99       |

|                                     |           |   |   |
|-------------------------------------|-----------|---|---|
|                                     | $F(1, 1)$ | = | . |
| <code>corr(u_i, Xb) = 0.0000</code> | Prob > F  | = | . |

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_r~e | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|-----------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| U_K_inflation   |             |                                   |       |       |                      |          |
| --.             | -.0003876   | .0002236                          | -1.73 | 0.333 | -.0032284            | .0024533 |
| L1.             | -.000301    | .0001286                          | -2.34 | 0.257 | -.0019347            | .0013328 |
| L2.             | -.000112    | .0000473                          | -2.37 | 0.255 | -.0007133            | .0004893 |
| L3.             | -.0000594   | .00023                            | -0.26 | 0.839 | -.0029819            | .0028633 |
| L4.             | -.0000785   | .0002373                          | -0.33 | 0.797 | -.0030935            | .0029365 |
| _cons           | .0275427    | .000386                           | 71.35 | 0.009 | .0226381             | .0324472 |
| sigma_u         | .0117044    |                                   |       |       |                      |          |
| sigma_e         | .01161931   |                                   |       |       |                      |          |
| rho             | .50364831   | (fraction of variance due to u_i) |       |       |                      |          |

```
50 . * test Euro inflation to the model
```

```
51 . xtreg Delinquency_rate L(3).Real_GDP_growth L(1).CPI_inflation_rate L(2).Unemployment_rate L(4).BBB_corporate_yield Market_Volatility_Index_Level_ L(3).Euro_area_r
> eal_GDP_growth L(1).Euro_area_inflation, fe vce(cluster entity_id)
```

|                                   |                  |   |     |
|-----------------------------------|------------------|---|-----|
| Fixed-effects (within) regression | Number of obs    | = | 198 |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2   |

|                  |                |
|------------------|----------------|
| R-squared:       | Obs per group: |
| Within = 0.7280  | min = 99       |
|                  | avg = 99.0     |
| Overall = 0.4817 | max = 99       |

|                        |          |   |   |
|------------------------|----------|---|---|
| corr(u_i, Xb) = 0.0000 | F(1, 1)  | = | . |
|                        | Prob > F | = | . |

(Std. err. adjusted for 2 clusters in **entity\_id**)

|                  | Robust      |           |   |      |                      |
|------------------|-------------|-----------|---|------|----------------------|
| Delinquency_rate | Coefficient | std. err. | t | P> t | [95% conf. interval] |
| Real_GDP_growth  |             |           |   |      |                      |

|                                |                     |           |                                   |         |       |           |           |
|--------------------------------|---------------------|-----------|-----------------------------------|---------|-------|-----------|-----------|
|                                | L3.                 | -.000327  | .0000146                          | -22.37  | 0.028 | -.0005127 | -.0001412 |
|                                | CPI_inflation_rate  |           |                                   |         |       |           |           |
|                                | L1.                 | -.0006217 | 2.19e-06                          | -284.29 | 0.002 | -.0006495 | -.0005939 |
|                                | Unemployment_rate   |           |                                   |         |       |           |           |
|                                | L2.                 | .0008812  | .0002854                          | 3.09    | 0.199 | -.0027447 | .0045071  |
|                                | BBB_corporate_yield |           |                                   |         |       |           |           |
|                                | L4.                 | .0057773  | .0008657                          | 6.67    | 0.095 | -.0052221 | .0167766  |
| Market_Volatility_Index_Level_ |                     | .0001078  | .0000128                          | 8.45    | 0.075 | -.0000544 | .0002699  |
| Euro_area_real_GDP_growth      |                     |           |                                   |         |       |           |           |
|                                | L3.                 | .0001697  | .0000264                          | 6.43    | 0.098 | -.0001654 | .0005048  |
| Euro_area_inflation            |                     |           |                                   |         |       |           |           |
|                                | L1.                 | .000737   | .0000143                          | 51.40   | 0.012 | .0005548  | .0009192  |
|                                | _cons               | -.0126611 | .0024973                          | -5.07   | 0.124 | -.0443917 | .0190695  |
|                                | sigma_u             | .0117044  |                                   |         |       |           |           |
|                                | sigma_e             | .00617976 |                                   |         |       |           |           |
|                                | rho                 | .78200184 | (fraction of variance due to u_i) |         |       |           |           |

```
52 . xtreg Delinquency_rate L(3).Real_GDP_growth L(1).CPI_inflation_rate L(2).Unemployment_rate L(4).BBB_corporate_yield Market_Volatility_Index_Level_ L(3).Euro_area_r
> eal_GDP_growth L(2).Euro_area_inflation, fe vce(cluster entity_id)
```

```

Fixed-effects (within) regression                               Number of obs   =       198
Group variable: entity_id                                   Number of groups =        2

R-squared:
    Within = 0.7239                                           Obs per group:
    Overall = 0.4790                                           min =           99
                                                                avg =          99.0
                                                                max =           99

corr(u_i, Xb) = 0.0000                                         F(1, 1)         =        .
                                                                Prob > F         =        .

                                                                (Std. err. adjusted for 2 clusters in entity_id)

```

| Delinquency_rate                                                   | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|--------------------------------------------------------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Real_GDP_growth<br>L3.                                             | -.0003545   | .0000177                          | -20.02 | 0.032 | -.0005795            | -.0001295 |
| CPI_inflation_rate<br>L1.                                          | -.0003328   | .0000512                          | -6.50  | 0.007 | -.0009828            | .0003173  |
| Unemployment_rate<br>L2.                                           | .0009081    | .0002465                          | 3.68   | 0.169 | -.0022238            | .00404    |
| BBB_corporate_yield<br>L4.                                         | .0057631    | .0008742                          | 6.59   | 0.096 | -.0053449            | .0168712  |
| Market_Volatility_Index_Level_<br>Euro_area_real_GDP_growth<br>L3. | .0001104    | .0000139                          | 7.95   | 0.000 | -.0000662            | .000287   |
| Euro_area_inflation<br>L2.                                         | .0002232    | .0000299                          | 7.48   | 0.005 | -.0001561            | .0006026  |
| _cons                                                              | .0003987    | .0001322                          | 3.02   | 0.204 | -.0012808            | .0020782  |
|                                                                    | -.012852    | .0029144                          | -4.41  | 0.142 | -.0498836            | .0241795  |
| sigma_u                                                            | .0117044    |                                   |        |       |                      |           |
| sigma_e                                                            | .0062262    |                                   |        |       |                      |           |
| rho                                                                | .77943866   | (fraction of variance due to u_i) |        |       |                      |           |

```
53 . xtreg Delinquency_rate L(3).Real_GDP_growth L(1).CPI_inflation_rate L(2).Unemployment_rate L(4).BBB_corporate_yield Market_Volatility_Index_Level_ L(1).Euro_area_i
> nflation, fe vce(cluster entity_id)
```

|                                   |                  |   |             |
|-----------------------------------|------------------|---|-------------|
| Fixed-effects (within) regression | Number of obs    | = | 198         |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2           |
| R-squared:                        | Obs per group:   |   |             |
| Within = <b>0.7261</b>            | min =            |   | <b>99</b>   |
|                                   | avg =            |   | <b>99.0</b> |
| Overall = <b>0.4805</b>           | max =            |   | <b>99</b>   |
|                                   | $F(1, 1)$        | = | .           |
| corr(u_i, Xb) = <b>-0.0000</b>    | Prob > F         | = | .           |

3/27/26, 9:06 PM

| Delinquency_rate                      | Coefficient | Robust<br>std. err.               | t       | P> t  | [95% conf. interval] |           |
|---------------------------------------|-------------|-----------------------------------|---------|-------|----------------------|-----------|
| Real_GDP_growth<br>L3.                | -.0001138   | .0000185                          | -6.16   | 0.103 | -.0003485            | .000121   |
| CPI_inflation_rate<br>L1.             | -.0006359   | 4.39e-06                          | -144.86 | 0.004 | -.0006917            | -.0005802 |
| Unemployment_rate<br>L2.              | .0008671    | .0002824                          | 3.07    | 0.200 | -.0027215            | .0044558  |
| BBB_corporate_yield<br>L4.            | .0057988    | .0008601                          | 6.74    | 0.094 | -.0051294            | .0167269  |
| Market_Volatility_Index_Level_<br>L1. | .0001121    | .0000134                          | 8.37    | 0.076 | -.0000581            | .0002823  |
| Euro_area_inflation<br>L1.            | .0007972    | .0000236                          | 33.73   | 0.019 | .0004969             | .0010975  |
| _cons                                 | -.0131448   | .0024157                          | -5.44   | 0.116 | -.0438394            | .0175498  |
| sigma_u                               | .0117044    |                                   |         |       |                      |           |
| sigma_e                               | .00618465   |                                   |         |       |                      |           |
| rho                                   | .78173213   | (fraction of variance due to u_i) |         |       |                      |           |

54 . \* test Japan inflation to the model

```
55 . xtreg Delinquency_rate L(3).Real_GDP_growth L(1).CPI_inflation_rate L(2).Unemployment_rate L(4).BBB_corporate_yield Market_Volatility_Index_Level_ L(3).Euro_area_r
> eal_GDP_growth Japan_inflation, fe vce(cluster entity_id)
```

```
Fixed-effects (within) regression      Number of obs   =    198
Group variable: entity_id             Number of groups =     2
```

|                  |                |
|------------------|----------------|
| R-squared:       | Obs per group: |
| Within = 0.7284  | min = 99       |
|                  | avg = 99.0     |
| Overall = 0.4820 | max = 99       |

|                        |                                   |   |   |
|------------------------|-----------------------------------|---|---|
| corr(u_i, Xb) = 0.0000 | $\frac{F(1, 1)}{\text{Prob} > F}$ | = | . |
|                        |                                   | = | . |

(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate                      | Coefficient            | Robust<br>std. err.               | t              | P> t           | [95% conf. interval]   |                      |
|---------------------------------------|------------------------|-----------------------------------|----------------|----------------|------------------------|----------------------|
| Real_GDP_growth<br>L3.                | -.0003255              | 9.87e-06                          | -32.97         | 0.019          | -.0004509              | -.0002001            |
| CPI_inflation_rate<br>L1.             | -.0000315              | .0000308                          | -1.02          | 0.492          | -.0004225              | .0003595             |
| Unemployment_rate<br>L2.              | .00061                 | .0002559                          | 2.38           | 0.253          | -.0026419              | .003862              |
| BBB_corporate_yield<br>L4.            | .0055659               | .0008924                          | 6.24           | 0.101          | -.0057726              | .0169043             |
| Market_Volatility_Index_Level_<br>L3. | .0000952               | .0000101                          | 9.39           | 0.068          | -.0000336              | .000224              |
| Euro_area_real_GDP_growth<br>L3.      | .0001769               | .0000196                          | 9.03           | 0.070          | -.0000719              | .0004257             |
| Japan_inflation<br>_cons              | -.0006954<br>-.0091752 | .0001039<br>.0028871              | -6.69<br>-3.18 | 0.094<br>0.194 | -.0020158<br>-.0458591 | .0006249<br>.0275088 |
| sigma_u                               | .0117044               |                                   |                |                |                        |                      |
| sigma_e                               | .00617519              |                                   |                |                |                        |                      |
| rho                                   | .78225407              | (fraction of variance due to u_i) |                |                |                        |                      |

```
56 . xtreg Delinquency_rate L(3).Real_GDP_growth L(1).CPI_inflation_rate L(2).Unemployment_rate L(4).BBB_corporate_yield Market_Volatility_Index_Level_ L(3).Euro_area_r
> eal GDP_growth L(1).Japan_inflation, fe vce(cluster entity_id)
```

```
Fixed-effects (within) regression      Number of obs   =    198
Group variable: entity_id             Number of groups =     2
```

|                 |                |
|-----------------|----------------|
| R-squared:      | Obs per group: |
| Within = 0.7445 | min = 99       |
|                 | avg = 99.0     |

Overall = 0.4927

max = 99

corr(u\_i, Xb) = -0.0000

$F(1, 1)$

=

.

Prob > F

=

.

(Std. err. adjusted for 2 clusters in entity\_id)

| Delinquency_rate                 | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|----------------------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Real_GDP_growth<br>L3.           | -.0003906   | .0000216                          | -18.08 | 0.035 | -.0006651            | -.0001161 |
| CPI_inflation_rate<br>L1.        | .0000574    | .0000618                          | 0.93   | 0.524 | -.0007275            | .0008422  |
| Unemployment_rate<br>L2.         | .0005053    | .00022                            | 2.30   | 0.261 | -.0022906            | .0033012  |
| BBB_corporate_yield<br>L4.       | .0054767    | .000924                           | 5.93   | 0.106 | -.0062638            | .0172172  |
| Market_Volatility_Index_Level_   | .0001256    | .0000155                          | 8.09   | 0.078 | -.0000715            | .0003227  |
| Euro_area_real_GDP_growth<br>L3. | .0002776    | .0000379                          | 7.32   | 0.086 | -.0002045            | .0007598  |
| Japan_inflation<br>L1.           | -.0011465   | .0002512                          | -4.56  | 0.137 | -.0043385            | .0020456  |
| _cons                            | -.0089537   | .0031083                          | -2.88  | 0.213 | -.0484486            | .0305411  |
| sigma_u                          | .0117044    | (fraction of variance due to u_i) |        |       |                      |           |
| sigma_e                          | .00598913   |                                   |        |       |                      |           |
| rho                              | .79249638   |                                   |        |       |                      |           |

57 . \* Japan inflation makes the model overfit.

58 . \* Do we need to include Euro inflation? Not really

59 . \* Euro Inflation does not influence US default directly

60 . \* test dependency of delinquency rate to EUR exchange rate

61 . xtreg Delinquency\_rate Euro\_area\_bilateral\_dollar\_excha, fe vce(cluster entity\_id)

Fixed-effects (within) regression

Group variable: entity\_id

R-squared:

Within = 0.0078

Overall = 0.0050

Number of obs = 206

Number of groups = 2

Obs per group:

min = 103

avg = 103.0

max = 103

corr(u\_i, Xb) = -0.0000

$F(1, 1)$

=

0.30

Prob > F

=

0.6827

(Std. err. adjusted for 2 clusters in entity\_id)

| Delinquency_rate                 | Coefficient | Robust<br>std. err.               | t    | P> t  | [95% conf. interval] |          |
|----------------------------------|-------------|-----------------------------------|------|-------|----------------------|----------|
| Euro_area_bilateral_dollar_excha | .0065152    | .0119724                          | 0.54 | 0.683 | -.145609             | .1586395 |
| _cons                            | .0176185    | .0142565                          | 1.24 | 0.433 | -.1635275            | .1987645 |
| sigma_u                          | .01188763   | (fraction of variance due to u_i) |      |       |                      |          |
| sigma_e                          | .0115116    |                                   |      |       |                      |          |
| rho                              | .51606633   |                                   |      |       |                      |          |

62 . xtreg Delinquency\_rate L(1/4).Euro\_area\_bilateral\_dollar\_excha, fe vce(cluster entity\_id)

Fixed-effects (within) regression

Group variable: entity\_id

R-squared:

Within = 0.0193

Overall = 0.0128

Number of obs = 198

Number of groups = 2

Obs per group:

min = 99

avg = 99.0

max = 99

corr(u\_i, Xb) = -0.0000

$F(1, 1)$

=

.

Prob > F

=

.

(Std. err. adjusted for 2 clusters in entity\_id)

| Delinquency_rate                 | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|----------------------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Euro_area_bilateral_dollar_excha |             |                                   |        |       |                      |           |
| L1.                              | .0228887    | .0106193                          | 2.16   | 0.277 | -.1120423            | .1578196  |
| L2.                              | -.0081335   | .0006242                          | -13.03 | 0.049 | -.0160642            | -.0002027 |
| L3.                              | -.0016845   | .0020337                          | -0.83  | 0.560 | -.0275248            | .0241558  |
| L4.                              | -.007471    | .0020223                          | -3.69  | 0.168 | -.0331661            | .0182242  |
| _cons                            | .0182887    | .0183202                          | 1.00   | 0.501 | -.2144909            | .2510684  |
| sigma_u                          | .0117044    |                                   |        |       |                      |           |
| sigma_e                          | .01164167   |                                   |        |       |                      |           |
| rho                              | .50268713   | (fraction of variance due to u_i) |        |       |                      |           |

63 . xtreg Delinquency\_rate Euro\_area\_bilateral\_dollar\_excha L(1/4).Euro\_area\_bilateral\_dollar\_excha, fe vce(cluster entity\_id)

|                                                          |                  |   |      |
|----------------------------------------------------------|------------------|---|------|
| Fixed-effects (within) regression                        | Number of obs    | = | 198  |
| Group variable: <b>entity_id</b>                         | Number of groups | = | 2    |
| R-squared:                                               | Obs per group:   |   |      |
| Within = <b>0.0378</b>                                   | min =            |   | 99   |
|                                                          | avg =            |   | 99.0 |
| Overall = <b>0.0250</b>                                  | max =            |   | 99   |
|                                                          | F(1, 1)          | = | .    |
| corr(u_i, Xb) = <b>0.0000</b>                            | Prob > F         | = | .    |
| (Std. err. adjusted for 2 clusters in <b>entity_id</b> ) |                  |   |      |

| Delinquency_rate                 | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |          |
|----------------------------------|-------------|-----------------------------------|--------|-------|----------------------|----------|
| Euro_area_bilateral_dollar_excha |             |                                   |        |       |                      |          |
| --.                              | .028158     | .0096558                          | 2.92   | 0.210 | -.0945307            | .1508468 |
| L1.                              | -.0034946   | .0015997                          | -2.18  | 0.273 | -.0238208            | .0168316 |
| L2.                              | -.0077123   | .0007702                          | -10.01 | 0.063 | -.0174989            | .0020744 |
| L3.                              | -.0018407   | .0019854                          | -0.93  | 0.524 | -.0270678            | .0233865 |
| L4.                              | -.0069803   | .0021958                          | -3.18  | 0.194 | -.03488              | .0209193 |
| _cons                            | .0151914    | .0194299                          | 0.78   | 0.578 | -.2316891            | .2620719 |
| sigma_u                          | .0117044    |                                   |        |       |                      |          |
| sigma_e                          | .01156192   |                                   |        |       |                      |          |
| rho                              | .50612359   | (fraction of variance due to u_i) |        |       |                      |          |

64 . \* test dependency of delinquency rate to Asia exchange rate

65 . xtreg Delinquency\_rate Developing\_Asia\_bilateral\_dollar, fe vce(cluster entity\_id)

|                                                          |                  |   |        |
|----------------------------------------------------------|------------------|---|--------|
| Fixed-effects (within) regression                        | Number of obs    | = | 206    |
| Group variable: <b>entity_id</b>                         | Number of groups | = | 2      |
| R-squared:                                               | Obs per group:   |   |        |
| Within = <b>0.0454</b>                                   | min =            |   | 103    |
|                                                          | avg =            |   | 103.0  |
| Overall = <b>0.0295</b>                                  | max =            |   | 103    |
|                                                          | F(1, 1)          | = | 23.63  |
| corr(u_i, Xb) = <b>0.0000</b>                            | Prob > F         | = | 0.1292 |
| (Std. err. adjusted for 2 clusters in <b>entity_id</b> ) |                  |   |        |

| Delinquency_rate                 | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|----------------------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Developing_Asia_bilateral_dollar |             |                                   |       |       |                      |          |
| _cons                            | .0003554    | .0000731                          | 4.86  | 0.129 | -.0005736            | .0012845 |
|                                  | -.0087252   | .0070148                          | -1.24 | 0.431 | -.0978572            | .0804068 |
| sigma_u                          | .01188763   |                                   |       |       |                      |          |
| sigma_e                          | .01129118   |                                   |       |       |                      |          |
| rho                              | .52571577   | (fraction of variance due to u_i) |       |       |                      |          |

66 . xtreg Delinquency\_rate L(1/4).Developing\_Asia\_bilateral\_dollar, fe vce(cluster entity\_id)

|                                   |                  |   |      |
|-----------------------------------|------------------|---|------|
| Fixed-effects (within) regression | Number of obs    | = | 198  |
| Group variable: <b>entity_id</b>  | Number of groups | = | 2    |
| R-squared:                        | Obs per group:   |   |      |
| Within = <b>0.0895</b>            | min =            |   | 99   |
|                                   | avg =            |   | 99.0 |

Overall = **0.0592**max = **99**corr(u\_i, Xb) = **0.0000**

$$\frac{F(1, 1)}{\text{Prob} > F} = .$$
(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate                 | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|----------------------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Developing_Asia_bilateral_dollar |             |                                   |       |       |                      |          |
| L1.                              | -.0003554   | .0001238                          | -2.87 | 0.213 | -.0019279            | .0012171 |
| L2.                              | .0002314    | .0000659                          | 3.51  | 0.177 | -.0006059            | .0010687 |
| L3.                              | .0002168    | .0000458                          | 4.73  | 0.133 | -.0003652            | .0007989 |
| L4.                              | .0004081    | .0001714                          | 2.38  | 0.253 | -.0017699            | .0025861 |
| _cons                            | -.022738    | .0061557                          | -3.69 | 0.168 | -.1009538            | .0554778 |
| sigma_u                          | .0117044    |                                   |       |       |                      |          |
| sigma_e                          | .01121737   |                                   |       |       |                      |          |
| rho                              | .52123815   | (fraction of variance due to u_i) |       |       |                      |          |

67 . xtreg Delinquency\_rate Developing\_Asia\_bilateral\_dollar L(1/4).Developing\_Asia\_bilateral\_dollar, fe vce(cluster entity\_id)

Fixed-effects (within) regression  
Group variable: **entity\_id**Number of obs = **198**  
Number of groups = **2**

R-squared:

Within = **0.1045**

Obs per group:

min = **99**  
avg = **99.0**  
max = **99**Overall = **0.0692**corr(u\_i, Xb) = **-0.0000**

$$\frac{F(1, 1)}{\text{Prob} > F} = .$$
(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate                 | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|----------------------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Developing_Asia_bilateral_dollar |             |                                   |       |       |                      |          |
| --.                              | -.0007382   | .0001526                          | -4.84 | 0.130 | -.0026771            | .0012006 |
| L1.                              | .0003527    | .0000223                          | 15.83 | 0.040 | .0000696             | .0006359 |
| L2.                              | .0002028    | .000072                           | 2.82  | 0.217 | -.0007119            | .0011175 |
| L3.                              | .0003369    | .0000211                          | 15.95 | 0.040 | .0000685             | .0006052 |
| L4.                              | .0003172    | .0001531                          | 2.07  | 0.286 | -.0016277            | .0022621 |
| _cons                            | -.0198915   | .0067601                          | -2.94 | 0.209 | -.1057864            | .0660033 |
| sigma_u                          | .0117044    |                                   |       |       |                      |          |
| sigma_e                          | .01115344   |                                   |       |       |                      |          |
| rho                              | .5240898    | (fraction of variance due to u_i) |       |       |                      |          |

68 . \* test dependency of delinquency rate to Japan exchange rate

69 . xtreg Delinquency\_rate Japan\_bilateral\_dollar\_exchange\_, fe vce(cluster entity\_id)

Fixed-effects (within) regression  
Group variable: **entity\_id**Number of obs = **206**  
Number of groups = **2**

R-squared:

Within = **0.0442**

Obs per group:

min = **103**  
avg = **103.0**  
max = **103**Overall = **0.0287**corr(u\_i, Xb) = **0.0000**

$$\frac{F(1, 1)}{\text{Prob} > F} = 48.54$$
(Std. err. adjusted for 2 clusters in **entity\_id**)

| Delinquency_rate                 | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|----------------------------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| Japan_bilateral_dollar_exchange_ |             |                                   |       |       |                      |          |
| _cons                            | -.0001346   | .0000193                          | -6.97 | 0.091 | -.00038              | .0001109 |
|                                  | .040486     | .0021687                          | 18.67 | 0.034 | .0129295             | .0680425 |
| sigma_u                          | .01188763   |                                   |       |       |                      |          |
| sigma_e                          | .01129842   |                                   |       |       |                      |          |
| rho                              | .52539602   | (fraction of variance due to u_i) |       |       |                      |          |

70 . xtreg Delinquency\_rate L(1/4).Japan\_bilateral\_dollar\_exchange\_, fe vce(cluster entity\_id)

```

Fixed-effects (within) regression      Number of obs   =      198
Group variable: entity_id           Number of groups =       2

R-squared:                            Obs per group:
    Within = 0.1014                  min =      99
    Overall = 0.0671                  avg =     99.0
                                      max =      99

corr(u_i, Xb) = -0.0000              F(1, 1)         =      .
                                      Prob > F          =      .

                                      (Std. err. adjusted for 2 clusters in entity_id)

```

| Delinquency_rate                 | Coefficient | Robust<br>std. err. | t     | P> t  | [95% conf. interval] |                                   |
|----------------------------------|-------------|---------------------|-------|-------|----------------------|-----------------------------------|
| Japan_bilateral_dollar_exchange_ |             |                     |       |       |                      |                                   |
| L1.                              | -.0004258   | .0000445            | -9.58 | 0.066 | -.0009908            | .0001391                          |
| L2.                              | .000049     | 2.07e-06            | 23.67 | 0.027 | .0000227             | .0000753                          |
| L3.                              | -.0000209   | 3.53e-06            | -5.94 | 0.106 | -.0000657            | .0000239                          |
| L4.                              | .0003822    | .0000376            | 10.16 | 0.062 | -.000096             | .0008605                          |
| _cons                            | .0273352    | .0006557            | 41.69 | 0.015 | .0190035             | .0356668                          |
| sigma_u                          | .0117044    |                     |       |       |                      |                                   |
| sigma_e                          | .01114408   |                     |       |       |                      |                                   |
| rho                              | .52450864   |                     |       |       |                      | (fraction of variance due to u_i) |

```
71 . xtreg Delinquency_rate Japan_bilateral_dollar_exchange_ L(1/4).Japan_bilateral_dollar_exchange_, fe vce(cluster entity_id)
```

```

Fixed-effects (within) regression      Number of obs   =      198
Group variable: entity_id           Number of groups =       2

R-squared:                            Obs per group:
    Within = 0.1439                  min =      99
    Overall = 0.0952                  avg =     99.0
                                      max =      99

corr(u_i, Xb) = 0.0000              F(1, 1)         =      .
                                      Prob > F          =      .

                                      (Std. err. adjusted for 2 clusters in entity_id)

```

| Delinquency_rate                 | Coefficient | Robust<br>std. err. | t      | P> t  | [95% conf. interval] |                                   |
|----------------------------------|-------------|---------------------|--------|-------|----------------------|-----------------------------------|
| Japan_bilateral_dollar_exchange_ |             |                     |        |       |                      |                                   |
| --.                              | -.0004008   | .0000466            | -8.61  | 0.074 | -.0009925            | .0001908                          |
| L1.                              | -.0000677   | 2.97e-06            | -22.79 | 0.028 | -.0001054            | -.0000299                         |
| L2.                              | .0000124    | 6.33e-06            | 1.96   | 0.300 | -.000068             | .0000928                          |
| L3.                              | .0001513    | .0000235            | 6.43   | 0.098 | -.0001479            | .0004504                          |
| L4.                              | .0002722    | .000025             | 10.91  | 0.058 | -.0000449            | .0005892                          |
| _cons                            | .0293603    | .0008927            | 32.89  | 0.019 | .0180176             | .040703                           |
| sigma_u                          | .0117044    |                     |        |       |                      |                                   |
| sigma_e                          | .01090574   |                     |        |       |                      |                                   |
| rho                              | .53527912   |                     |        |       |                      | (fraction of variance due to u_i) |

```
72 . * test dependency of delinquency rate to UK exchange rate
```

```
73 . xtreg Delinquency_rate U_K_bilateral_dollar_exchange_r, fe vce(cluster entity_id)
```

```

Fixed-effects (within) regression      Number of obs   =      206
Group variable: entity_id           Number of groups =       2

R-squared:                            Obs per group:
    Within = 0.1116                  min =     103
    Overall = 0.0726                  avg =    103.0
                                      max =     103

corr(u_i, Xb) = 0.0000              F(1, 1)         =      2.78
                                      Prob > F          =     0.3442

                                      (Std. err. adjusted for 2 clusters in entity_id)

```

| Delinquency_rate                | Coefficient | Robust<br>std. err. | t     | P> t  | [95% conf. interval] |          |
|---------------------------------|-------------|---------------------|-------|-------|----------------------|----------|
| U_K_bilateral_dollar_exchange_r | .0172004    | .0103253            | 1.67  | 0.344 | -.1139946            | .1483953 |
| _cons                           | -.0007501   | .0156838            | -0.05 | 0.970 | -.2000314            | .1985311 |
| sigma_u                         | .01188763   |                     |       |       |                      |          |



```

sigma_e .01089273
rho .5435906 (fraction of variance due to u_i)

```

```
74 . xtreg Delinquency_rate L(1/4).U_K_bilateral_dollar_exchange_r, fe vce(cluster entity_id)
```

```

Fixed-effects (within) regression      Number of obs   =      198
Group variable: entity_id             Number of groups =       2

R-squared:                             Obs per group:
    Within = 0.1512                     min =      99
    Overall = 0.1001                     avg =     99.0
                                         max =      99

corr(u_i, Xb) = 0.0000                  F(1, 1)         =      .
                                         Prob > F         =      .

                                         (Std. err. adjusted for 2 clusters in entity_id)

```

| Delinquency_rate                | Coefficient | Robust<br>std. err. | t     | P> t  | [95% conf. interval] |                                   |
|---------------------------------|-------------|---------------------|-------|-------|----------------------|-----------------------------------|
| U_K_bilateral_dollar_exchange_r |             |                     |       |       |                      |                                   |
| L1.                             | .006877     | .000872             | 7.89  | 0.000 | -.0042029            | .0179568                          |
| L2.                             | -.0054048   | .0024169            | -2.24 | 0.268 | -.0361149            | .0253054                          |
| L3.                             | -.0101936   | .0014256            | -7.15 | 0.000 | -.0283075            | .0079202                          |
| L4.                             | .0285636    | .0077308            | 3.69  | 0.168 | -.0696654            | .1267925                          |
| _cons                           | -.0052526   | .0189824            | -0.28 | 0.828 | -.2464464            | .2359411                          |
| sigma_u                         | .0117044    |                     |       |       |                      |                                   |
| sigma_e                         | .01083041   |                     |       |       |                      |                                   |
| rho                             | .53872571   |                     |       |       |                      | (fraction of variance due to u_i) |

```
75 . xtreg Delinquency_rate U_K_bilateral_dollar_exchange_r L(1/4).U_K_bilateral_dollar_exchange_r, fe vce(cluster entity_id)
```

```

Fixed-effects (within) regression      Number of obs   =      198
Group variable: entity_id             Number of groups =       2

R-squared:                             Obs per group:
    Within = 0.1630                     min =      99
    Overall = 0.1079                     avg =     99.0
                                         max =      99

corr(u_i, Xb) = -0.0000                 F(1, 1)         =      .
                                         Prob > F         =      .

                                         (Std. err. adjusted for 2 clusters in entity_id)

```

| Delinquency_rate                | Coefficient | Robust<br>std. err. | t      | P> t  | [95% conf. interval] |                                   |
|---------------------------------|-------------|---------------------|--------|-------|----------------------|-----------------------------------|
| U_K_bilateral_dollar_exchange_r |             |                     |        |       |                      |                                   |
| --.                             | .0182051    | .0023312            | 7.81   | 0.001 | -.0114156            | .0478258                          |
| L1.                             | -.013549    | .0017413            | -7.78  | 0.001 | -.0356743            | .0085764                          |
| L2.                             | .0001993    | .0031408            | 0.06   | 0.960 | -.0397088            | .0401075                          |
| L3.                             | -.0133727   | .0010222            | -13.08 | 0.049 | -.0263611            | -.0003843                         |
| L4.                             | .029236     | .007837             | 3.73   | 0.167 | -.0703425            | .1288146                          |
| _cons                           | -.0065575   | .0191988            | -0.34  | 0.790 | -.2505017            | .2373867                          |
| sigma_u                         | .0117044    |                     |        |       |                      |                                   |
| sigma_e                         | .01078309   |                     |        |       |                      |                                   |
| rho                             | .54090155   |                     |        |       |                      | (fraction of variance due to u_i) |

```
76 . * Exchange rate is significant at lag 2 but exchange rate is too noisy for the model
```

```
77 . * Euro GDP enough to explain the global market cycle
```

```
78 . * Final Model
```

```
79 . xtreg Delinquency_rate L(3).Real_GDP_growth L(1).CPI_inflation_rate L(2).Unemployment_rate L(4).BBB_corporate_yield Market_Volatility_Index__Level_ L(3).Euro_area_r
> eal_GDP_growth, fe vce(cluster entity_id)
```

```

Fixed-effects (within) regression      Number of obs   =      198
Group variable: entity_id             Number of groups =       2

R-squared:                             Obs per group:
    Within = 0.7198                     min =      99
    Overall = 0.4763                     avg =     99.0
                                         max =      99

                                         F(1, 1)         =      .

```

corr(u\_i, Xb) = 0.0000

Prob > F = .

(Std. err. adjusted for 2 clusters in entity\_id)

| Delinquency_rate                 | Coefficient | Robust<br>std. err.               | t      | P> t  | [95% conf. interval] |           |
|----------------------------------|-------------|-----------------------------------|--------|-------|----------------------|-----------|
| Real_GDP_growth<br>L3.           | -.0003618   | .0000153                          | -23.72 | 0.027 | -.0005556            | -.0001679 |
| CPI_inflation_rate<br>L1.        | -.0001968   | 6.06e-06                          | -32.44 | 0.020 | -.0002738            | -.0001197 |
| Unemployment_rate<br>L2.         | .0007959    | .000283                           | 2.81   | 0.217 | -.0027995            | .0043912  |
| BBB_corporate_yield<br>L4.       | .0057395    | .0008641                          | 6.64   | 0.095 | -.0052403            | .0167194  |
| Market_Volatility_Index__Level_  | .0001135    | .0000128                          | 8.84   | 0.072 | -.0000496            | .0002767  |
| Euro_area_real_GDP_growth<br>L3. | .0002302    | .0000275                          | 8.38   | 0.076 | -.0001189            | .0005793  |
| _cons                            | -.0116528   | .0025103                          | -4.64  | 0.135 | -.0435489            | .0202433  |
| sigma_u                          | .0117044    |                                   |        |       |                      |           |
| sigma_e                          | .00625544   |                                   |        |       |                      |           |
| rho                              | .77782332   | (fraction of variance due to u_i) |        |       |                      |           |